

Determinants of Co-existence of Undernutrition and Anemia among Children aged 6-59 Months in Nepal: Evidence from the 2022 Nepal Demographic and Health Survey

--Manuscript Draft--

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Full Title:	Determinants of Co-existence of Undernutrition and Anemia among Children aged 6-59 Months in Nepal: Evidence from the 2022 Nepal Demographic and Health Survey
Short Title:	Factors Associated with Undernutrition, Anemia and Their Coexistence Among 6-59 Months Children in Nepal
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Keywords:	Undernutrition; Anemia; Co-existence; 6-59-months children; Nepal; NDHS 2022
Abstract:	Introduction: Undernutrition and anemia among children aged 6 to 59 months are significant public health issues in developing countries like Nepal. The co-existence of these conditions impact childhood development. This study aimed to determine the prevalence of undernutrition and anemia, assess their co-existence, and identify contributing factors among children aged 6-59 months in Nepal. Methods: We conducted a secondary analysis using data from 2022 Nepal Demographic and Health Survey (NDHS 2022), a nationally representative cross-sectional survey. The outcome variables were undernutrition, anemia, and their co-existence. Children aged 6-59 months were considered undernutrition if they exhibited either stunting, wasting, underweight or any combination of these conditions. Anemia was defined as hemoglobin levels <11.0 gm/dL (adjusted for altitude). We applied multivariable multinomial logistic regressions to determine factors associated with co-existence, and multivariable logistic regression to assess factors associated with undernutrition and anemia separately. We presented the results from the regression analysis using adjusted odds-ratio (aOR) and 95% confidence intervals (CI). Results: Among 2395 children, the weighted prevalence of undernutrition was 33.5% and of anemia was 43.4%, and co-existence of undernutrition and anemia was 16.0%. Children from the richest wealth quintile, whose mothers had at least secondary education, and those whose mother participated in household decision-making had 53% (aOR: 0.47; 95%CI: 0.26 to 0.86), 48% (aOR:0.52; 95%CI: 0.32 to 0.93), and 34% (aOR: 0.66; 95%CI: 0.47 to 0.94) lower odds of experiencing undernutrition and anemia co-existence compared to their counterparts. Children with malnourished mothers had 80% (aOR=1.80; 95% CI:1.20 to 2.70) higher odds of co-existence compared to their counterparts. Conclusion: The prevalence of undernutrition, anemia, and their co-existence among children was high in Nepal. Interventions that improve mother's education, strengthen their roles in family, and enhance household financial condition of household are essential to reduce these conditions and improve children's nutritional status.
Order of Authors:	Bikram Adhikari Biraj Neupane Jessica Rice Niharika Jha Kajol Dahal Parash Mani Sapkota Archana Shrestha Yu Jiang Xinhua Yu
Opposed Reviewers:	

Response to Reviewers:

Dear reviewers and editor,
Thank you for your time and effort to review and provide insightful and constructive comments on our research paper. We have addressed the comments point-by-point as follows:

Reviewer #1:

1. In the methods, authors state "we will analyze the data of 2,395 children from the NDHS 2022 dataset". However, the Abstract and the Results show 2,335. This discrepancy must be clarified. The authors should state the initial sample size and provide a clear explanation for any exclusions.

Response: Both numbers are corrected, 2335 is weighted frequency and 2395 is unweighted frequency. We have corrected and specified in the current version of the manuscript. In line 36, line 139, and table 2 (2nd and 3rd row) are now clarified the weighted and unweighted frequencies.

2. In Table 4, in the factors associated with anemia, the authors include "Nutritional status of child" as an independent variable. Given the cross-sectional nature of the data and the study's aim to assess coexistence, this approach may be problematic as it implies a causal direction that cannot be established and introduces potential endogeneity... The association between undernutrition and anemia is more appropriately explored through the multinomial logistic regression model (Table 5). In my opinion, this variable should be removed from the binary logistic regression model for anemia to avoid confusion and circular reasoning.

Response: Thank you for feedback, I agree with your comment, and we have removed under-nutrition and anemia as independent variables and treated them as dependent variables (Table 4, 5, and 6).

3. The manuscript uses the term "adjusted odds-like ratio (aOLR)". To the best of my knowledge, the output of a multinomial logistic regression model is typically referred to as a relative risk ratio (RRR) or, if presented as odds, as an odds ratio (OR). Please consider using a standard term and define it clearly. For example, "adjusted odds ratios (aOR) from the multinomial logistic regression model, with 'Normal' as the reference outcome category."

Response: The previous term was taken from an old textbook. Now we use an adjusted odds ratio (aOR) term for the multinomial logistic regression model to be consistent with the current terminology. The aOLR was now replaced with aOR throughout the paper.

4. The abstract states that children with malnourished mothers had a "90% (95% CI:1.24 to 2.90) higher likelihood" of coexistence. The CI corresponds to an OR of 1.90 (as shown in Table 5, which represents a 90% increase in odds, not likelihood. This distinction is important, and the language should be precise throughout the manuscript. The abstract should be revised to state: "...children with malnourished mothers had 90% higher odds (aOR: 1.90; 95% CI: 1.24 to 2.90) of..."

Response: Thank you for the feedback. We have corrected this throughout the manuscript. In line 41, Line 290-301, and other lines, we corrected sentences as per your comment.

5. Please correct table numbers, as they should be sequential (there are two tables numbered 1); perhaps using a section numbering?

Response: Thank you for this feedback. We have corrected table numbers throughout the paper.

Minor Comments

5. In the abstract, I suggest breaking down the results section into two or three separate sentences, for improved readability.

Response: The sentence (line 36-42) is split into 2 sentences.

6. In the introduction, the rationale for the study is clear. However, the statement that "there are limited studies exploring factors associated with the co-existence of undernutrition and anemia among U-5 children in Nepal" could be strengthened by briefly mentioning what the few existing studies (if any) have found or what specific gaps remain.

	Response: Few existing studies and their findings have been mentioned in the introduction along with the specific gap remaining. 7. In the methods, the definition of "undernutrition" is based on the Composite Index of Anthropometric Failure (CIAF) concept. It would be beneficial to explicitly name this index, as it is a well-established term and is supported by the cited reference. Response: Thanks. We have clarified this term. 8. The authors state they removed the 'province' variable from regression models due to a variance inflation factor (VIF) > 2. A threshold of 2 is quite conservative; VIF thresholds of 5 or 10 are more commonly used. Please provide justification for this choice. Response: Th VIF for the province variable was 13.31, so it was removed from the model based on either criterion. Note that the information of provinces is captured by the region and place of residence variable, i.e., these two variables were highly collinear. 9. The manuscript should explicitly state how missing data was handled in the statistical analysis. Several variables in Table 1 (e.g., father's education, mother's age, parity) have a substantial number of missing values. Were these cases excluded via listwise deletion for each model? This should be clarified in Statistical Analysis subsection. Response: They were excluded during analysis. Now it is mentioned in the statistical analysis subsection 10. The manuscript would benefit from a thorough proofread to correct minor grammatical errors and typos. For example: "presence anemia in children" (pg. 18, at the bottom) should be "presence of anemia"; "...those whose mothers who attained at least secondary level education, and those whose mothers participated..." is repetitive. Response: Thank you for your feedback. We wrote this manuscript without relying on any AI tools for grammatic check. We have thoroughly proofread the manuscript manually and corrected all the grammatical errors seen in the paper. Reviewer #2: The topic presented in the manuscript is highly relevant for public health in developing countries. Please consider making the title more specific to the scope and methods of your research. The definition of undernutrition is vague: "Stunting, wasting, underweight or any combination" — does this mean a child is considered undernourished if they have any one of the three? Or only if they have multiple? Response: The child was considered to have undernutrition if they had any one of the three conditions. This statement is clarified in line 163, page 6. Odds-like ratio" (aOLR) is uncommon and may confuse readers. Standardize with "adjusted odds ratio (aOR)" unless you are referring to a specific statistical nuance that requires clarification. Response: Thanks. This has been corrected throughout. Review grammatical issues and ambiguity of same phrases. Some sentences are wordy or unclear. Response: we have proofread the manuscript again and corrected all recognized grammatic errors. Contradictory finding needs explanation: You report that anemia and undernutrition were not associated with each other, yet some children had both. This warrants further discussion. If they co-exist at 16%, why is the statistical association not significant? Response: Thank you for your comment. We hypothesized the existence of an association between undernutrition and anemia after multivariate adjustment. However, our data did not support this hypothesis, thus the multivariable analyses (Table 5 and 6) were conducted with undernutrition and anemia separately. The crude
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association between undernutrition and anemia may be due to confounding factors such as socioeconomic status and regions included in the models. We also add this point in the discussion section.

In conclusions “Targeted interventions” is vague. What kind of interventions? Nutrition-specific? Maternal education? Social protection? Consider suggesting specific policies or areas for programmatic focus.

Response: Thank you for this comment. We have added specific interventions suggesting specific policies and areas of programmatic focus. That is, policies will be tailored to the specific needs in regions and based on the socioeconomic status of the people.

Reviewer #3: This is an interesting and well-written paper on a relevant topic. Please find below some issues which should be addressed to further improve the manuscript.

General remarks:

- The authors might re-check their article according to the STROBE checklist. For example, the study’s design should be indicated with a commonly used term in the title or the abstract.

Response: We have clarified in the abstract that this is a secondary data analysis of a cross-sectional survey.

- In the spirit of Open and Reproducible Science, the analysis code should be made available in an online repository together with a data dictionary, and the respective URL should be mentioned in the Methods section.

Response: The data and analysis code are accessible in the GitHub <https://github.com/Adrisyaone/Undernutritiona-and-anemia>

Detailed points:

- The title says this was a secondary analysis of the NDHS data. However, this term is not mentioned in the main text. It would therefore be helpful to give some context to the NDHS study, e.g. what its original aim was, how many participants were included etc.

Response: We have added original aim and number of participants for the original NDHS survey (line 127, page 4).

- Abstract: "were ... lower likelihood" is grammatically not correct. Please revise.

Response: Thank you for your feedback. It is now corrected as “lower odds”.

- Is "NDHS" the correct abbreviation for "Nepal Health Demographic Survey"? "NHDS" would seem more appropriate and is also mentioned like this once in the Methods section.

Response: NDHS is Nepal Demographic and Health Survey, so NDHS is the common abbreviation.

- The description of Nepal as a country might fit better into the Introduction. It would be helpful to the Reader to put the HDI values into context, e.g. compare them with those of the highest and least developed countries.

Response: Thanks. We have clarified this point in the method section regarding Nepal’s HDI and Global Hunger Index.

- "In this study, we will analyze the data of 2,395 children from the NDHS 2022 dataset." It should be mentioned for clarity that all these children were U-5. Were these children all U-5 children included in the DHS 2022 data? If not, by which criteria were the children selected for analysis?

Response: In this study all Under-5 children aged 6 months and above were taken for analysis. Now, we have mentioned which subsets of under 5 children are included for analysis.

- Please add a paragraph explaining how the data were collected, e.g. how height, weight and blood levels were in children and mothers measured, by whom (e.g. trained physicians) and in which setting (at school, in the children's homes...)? How were the remaining variables from table 1 and the potential confounders measured?

Response: we added brief information on these measures (Table 1) and referred them to the official documents of NDHS.

- I think "mother's nutritional status" is not a suitable word for describing their BMI category.

Response: it is changed to "Mother's BMI" throughout the manuscript

- Which population was used as reference to calculate z-scores of heights and weight?

Response: They were internally calculated (based on the overall population) in NDHS 2022.

- "province variable" is not a common statistical term. Please revise.

Response: Thank you. It is revised.

- "10.5% were 6-12 months old": Table 1 says 10.4%.

Response: Thank you. It is corrected

- Please note that there are two tables labelled as table 1.

Response: all tables are re-labeled and correctly numbered.

- The VENN diagram in Figure 1 is quite nice, but I think the information in Figure 2 might better be presented in a table, together with exact numbers (not only %). In Figure 2, it is not clear what the percentages in the upper right of each of the four panels indicate.

Response: The figure 1 is splitted into multiple Venn diagram based on the age group of children.

- Also in table 2, exact numbers should be mentioned in addition to %.

Response: Currently, the overall weighted N and unweighted N is kept in the table. This may make the table easy to read and the exact numbers could be computed in ad hoc.

- Figure 3: There should be no interpretation in the figure-legend, so the last sentence in the legend should be omitted. Instead, the meaning of the vertical bars (95% CIs?) should be explained.

Response: Thanks. Fixed. We explained the figure in the notes and text instead of legend.

- Were the analyses presented in tables 3 and 4 mutually adjusted for each other variable shown in the table? If not, why not?

Response: No, they were not mutually adjusted for each other. Both undernutrition and anemia were treated as dependent variables. The tables were revised as Table 5 and 6.

- "The child whose mother was thin" is awkward English wording and should be revised.

Response: The statement was rephrased as "The child whose mother was classified as thin (BMI<18 kg/m2)....".

- The second p-value column is incorrectly assigned to "Coexistence vs Normal".

Response: This issue with the table is corrected.

- In general, I would suggest removing all p-values from the tables, as the 95% CIs already contain information on whether an association was statistically significant or not.

Response: We agree with your statement, but some readers prefer p-values, thus we

	kept them in the table. - Discussion, first sentence: The prevalence of undernutrition should also be explicitly mentioned in the results section. The anemia prevalence of 43.5% is reported as 43.4% on other places in the manuscript. Response: it was a typo which is now corrected - "...which aligns with the rates observed in other LMICs. For instance, a study in Ethiopia reported a higher prevalence..." These two sentences do not fit together in this way. Response: Thank you for the feedback, we have changed the sentences and write in this way: A study in Ethiopia reported a higher prevalence of undernutrition among preschool children (50.8%) and children aged 6-59 months (57.3%) [32] reported higher prevalence of undernutrition compared to the findings from our study where we found prevalence of undernutrition among 6-59 months children in Nepal to be 33.5%. - "The decline in anemia prevalence in Nepal...": Does this mean this prevalence was higher previously? If so, a reference should be given which supports this statement. Response: Thank you, I've added reference to the statement. - "...our findings from Nepal, a low-middle-income country," This does not need to be repeated again. Response: Corrected - "The higher prevalence of co-existence of undernutrition and anemia in our study...": Higher compared to what? Response: The term higher was changed to high and this sentence wants to say 16% is high. - "Our study suggests that interventions that address the determinants of undernutrition should not be assumed to address undernutrition." I do not understand this sentence. Should the second "undernutrition" rather say "anemia"? Response: Thank you for the feedback. The last word should be anemia but because of typo, this sentence is not clear I've corrected this sentence. - "The children from the richest wealth quintile have lower odds..." Throughout the manuscript, when relating to results from this study, past tense should be used. Response: Corrected - How were non-response rates addressed, and how high were these rates (overall and for U-5 children)? Response: This is the complex survey analysis, which addresses the non-response rates and allows for generalization throughout the population. The response rates were higher which was >95%. - As a further limitation, the authors should discuss that the CIs between children's nutritional status and anemia were quite large, indicating that the statistical power / sample size was limited with respect to these associations. Response: Thank you for the constructive feedback, we have added this to the limitation section.
Additional Information:	
Question	Response
Financial Disclosure Enter a financial disclosure statement that describes the sources of funding for the	The author(s) received no specific funding for this work.

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We received permission from the official website of "the DHS program" (<https://www.dhsprogram.com>) to download and use (submitted on: 07/25/2023 and approved on 07/26/2023) NDHS 2022 dataset [21]. NHDS 2022 obtained ethical approval from the institutional review board of ICF International, United States of America (Reference number: 180657.0.001.NP.DHS.01, Date: 28th April 2022). and the ethical review board of Nepal Health Research Council (Reference number: 678, Date: 30th September 2021).

Format for specific study types

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- Give the name of the institutional review board or ethics committee that approved the study
- Include the approval number and/or a statement indicating approval of this research
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Animal Research (involving vertebrate animals, embryos or tissues)

- Provide the name of the Institutional Animal Care and Use Committee (IACUC) or other relevant ethics board that reviewed the study protocol, and indicate whether they approved this research or granted a formal waiver of ethical approval
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Data cannot be shared publicly because of [XXX]. Data are available from the XXX Institutional Data Access / Ethics Committee (contact via XXX) for researchers who meet the criteria for access to confidential data.

The data underlying the results presented in the study are available from (include the name of the third party

It is available in the public repository with the link:
https://dhsprogram.com/data/dataset/Nepal_Standard-DHS_2011.cfm?flag=1

<i>and contact information or URL).</i> • This text is appropriate if the data are owned by a third party and authors do not have permission to share the data. * typeset	
Additional data availability information:	

17th March 2025

To,
The editorial team
Plos One

We are pleased to submit our manuscript entitled “**Factors Associated with Undernutrition, Anemia and Their Coexistence Among 6-59 Months Children in Nepal: A Secondary Analysis of the Nepal Demographic and Health Survey-2022**” and supporting materials for your consideration.

We believe this paper particularly interesting because we analyzed secondary data from the nationally representative cross-sectional survey, Nepal Demographic and Health Survey 2022 (NDHS-2022) to to determine the prevalence of undernutrition, anemia, and their co-existence in Nepal, as well as determine their contributing factors among children aged 6-59 months.

We confirm that this manuscript is original and has not been published elsewhere or is under consideration for publication elsewhere. We have no conflicts of interest to disclose. Please address all correspondence concerning this manuscript to: bikram.adhikariadhitya@gmail.com

Sincerely,
Bikram Adhikari
University of Memphis
Memphis, TN, USA

Determinants of Co-existence of Undernutrition and Anemia among Children aged 6-59 Months in Nepal: Evidence from the 2022 Nepal Demographic and Health Survey

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Word count: 5103

Number of tables: 7

Number of figures: 4

Abstract

Introduction: Undernutrition and anemia among children aged 6 to 59 months are significant public health issues in developing countries like Nepal. The co-existence of these conditions impact childhood development. This study aimed to determine the prevalence of undernutrition and anemia, assess their co-existence, and identify contributing factors among children aged 6-59 months in Nepal.

Methods: We conducted a secondary analysis using data from 2022 Nepal Demographic and Health Survey (NDHS 2022), a nationally representative cross-sectional survey. The outcome variables were undernutrition, anemia, and their co-existence. Children aged 6-59 months were considered undernutrition if they exhibited either stunting, wasting, underweight or any combination of these conditions. Anemia was defined as hemoglobin levels <11.0 gm/dL (adjusted for altitude). We applied multivariable multinomial logistic regressions to determine factors associated with co-existence, and multivariable logistic regression to assess factors associated with undernutrition and anemia separately. We presented the results from the regression analysis using adjusted odds-ratio (aOR) and 95% confidence intervals (CI).

Results: Among 2395 children, the weighted prevalence of undernutrition was 33.5% and of anemia was 43.4%, and co-existence of undernutrition and anemia was 16.0%. Children from the richest wealth quintile, whose mothers had at least secondary education, and those whose mother participated in household decision-making had 53% (aOR: 0.47; 95%CI: 0.26 to 0.86), 48% (aOR:0.52; 95%CI: 0.32 to 0.93), and 34% (aOR: 0.66; 95%CI: 0.47 to 0.94) lower odds of experiencing undernutrition and anemia co-existence compared to their counterparts. Children with malnourished mothers had 80% (aOR=1.80; 95% CI:1.20 to 2.70) higher odds of co-existence compared to their counterparts.

Conclusion: The prevalence of undernutrition, anemia, and their co-existence among children was high in Nepal. Interventions that improve mother's education, strengthen their roles in family, and enhance household financial condition of household are essential to reduce these conditions and improve children's nutritional status.

Keywords: undernutrition; anemia; co-existence; 6-59-months children; Nepal; NDHS 2022

Introduction

Undernutrition including stunting, wasting, and underweight is a major public health challenge among children under 5 years (U-5) of age globally. This issue is particularly pronounced in developing countries like Nepal [1–4]. There are three different causes of undernutrition – immediate, underlying, and basic: The immediate causes include inadequate diet intake and repeated illnesses; The underlying causes include suboptimal child feeding and caregiver practices from mothers, lack of access to healthcare facilities, poverty, and food insecurity; The basic causes include limited information, political and economic insecurity, gender inequality, and the occurrence of natural disasters [5].

Different indexes of undernutrition measure different underlying development status. Stunting indicates chronic malnutrition reflecting long-term nutrient deficiencies and is linked with delayed motor development and impaired cognitive development [6]. Wasting indicates acute malnutrition associated with recent severe food shortages or illnesses leading to weight loss and is a strong predictor of mortality [6]. Underweight combines information about linear growth obstruction and weight for length or height [7].

Anemia, particularly iron deficiency anemia, is another common and pressing public health concern among children U-5 in low-, middle- and high-income countries. It results from poor nutrition and has severe adverse health consequences including impaired cognitive development, impaired immunity, disability, and increased risk of morbidity and mortality [8–10].

In 2022, the global prevalence of stunting among children under five years of age was 22.3%, while in South Asia, it reached 31.8%. Additionally, the global prevalence of wasting was at 6.8%, with South Asia reporting a prevalence of 14.8% [11]. In 2019, global anemia prevalence was 39.8% [12]. In Nepal, the prevalence of stunting, and wasting among children under 5 was 25% and 8% respectively, and the prevalence of anemia was 43% among children 6-59 months old [13].

The co-existence of undernutrition and anemia increases the risk of childhood morbidity and mortality [14–16]. Almost fifty percent of deaths among children under the age of five are associated with undernutrition, which predominantly occurs in low- and middle-income countries (LMICs). Notably, 88% of these countries (124 out of 141) experience multiple forms of malnutrition [17]. The developmental, economic, social, and medical impacts of undernutrition and anemia among children are serious and perpetuating, for individuals and their families, communities, and countries [7].

Furthermore, anemia and undernutrition are both concentrated in socioeconomically disadvantaged groups, which share numerous multifaceted causes and complex interactions between diet, transmissible illnesses, inadequate care and unhealthy household environments that adversely affect the cognitive development and physical well-being of children [18–20]. For instance, lower maternal

education levels and socioeconomic status of households are linked with higher rates of malnutrition among children [18].

Though the concern on nutrition should be explored beyond any single form of malnutrition to the co-existence of multiple forms, most studies in Nepal have focused on undernutrition and anemia separately. A few studies that have examined both conditions together primarily reported that children from poorer households, mothers with low education levels, and those with inadequate dietary diversity were more likely to experience concurrent undernutrition and anemia [21,22]. Even when both malnutrition and anemia were considered, anemia was often treated merely as a correlating factor or predictor rather than as part of a combined outcome reflecting their simultaneous occurrences. Moreover, there remains a gap in understanding how maternal factors (such as nutritional status and exposure to health information) and household-level socioeconomic characteristics interact to influence the co-existence of undernutrition and anemia among children under five in Nepal.

Therefore, this study aims to ascertain the prevalence of stunting, wasting, and underweight, as well as their co-occurrence, and the co-existence of undernutrition and anemia in Nepal. It also assesses the association of undernutrition, anemia, and co-existence of undernutrition with household economic status, maternal education and nutritional status, and exposure to televised or broadcasted health programs among children aged 6-59 months in Nepal. By identifying these factors, this study helps inform the development of targeted interventions and policies designed to mitigate malnutrition within this vulnerable population, thereby enhancing child health outcomes and alleviating the burden of malnutrition in Nepal.

Methods

Data source

In this study, we analyzed data from the Nepal Health Demographic Survey (NDHS) conducted in 2022 [23]. NDHS 2022 is the nationally representative survey implemented by New ERA under the aegis of the Ministry of Health and Population (MoHP) with the technical support of ICF International and funding from the United States Agency for International Development (USAID) [13]. This survey was conducted to provide estimates of key demographic and health indicators at the national and provincial levels. This study included 13,786 households, from which information on 14,845 women aged 15–49 years and 4,913 men aged 15–49 years and 2,643 children under 5 years.

Ethical approval

We received permission from the official website of “the DHS program” (<https://www.dhsprogram.com>) to download and use (submitted on: 07/25/2023 and approved on 07/26/2023) NDHS 2022 dataset [23]. NDHS 2022 obtained ethical approval from the institutional review board of ICF International, United States of America (Reference number: 180657.0.001.NP.DHS.01, Date: 28th April 2022). and the ethical review board of Nepal Health Research Council (Reference number: 678, Date: 30th September 2021) [13]. In the NDHS 2022, informed consent was taken from the participants before enrolling them into the study.

Study setting

This study used nationally representative data from Nepal, a landlocked country located in Southeast Asia with an area of 147, 516 km² [24]. It has seven administrative provinces, within which lies 753 municipalities (6 metropolitan cities, 11 sub-metropolitan cities, 276 urban municipalities, 460 rural municipalities) [24]. Nepal has three ecological belts- Mountain, Hill, and Terai. Based on the 2021 Census, the total population of Nepal was 29,164,578 of which 14,911,027 (51.1 %) were females and 14,253,551 (48.9 %) were males [24]. Based on the Nepal Human Development Report 2020, Nepal was ranked 147th based on the human development index (HDI) with an overall HDI of 0.587 and rural and urban parts of Nepal of 0.647 and 0.561 respectively. [25]. According to the Global Hunger Index, Nepal has a ranking of 19.5, implying moderate food insecurity and undernourishment risk, [26].

Sample and sampling

The NDHS 2022 used a two-stage stratified cluster sampling of households and stratification was achieved based on rural and urban settings. In the first stage of sampling, Primary Sampling Units (PSUs) were nominated by probability proportional to size followed by a systematic selection of households from individual PSUs during the second stage of sampling. In this study, we will analyze the data of 2,395 children aged 6 months to 5 years using NDHS 2022 dataset.

Measures

Outcome variables

The primary outcome was “Coexistence of undernutrition (stunting, wasting or underweight) and anemia”. Anemia was assessed using altitude-adjusted hemoglobin levels and a child was considered anemic if the level was less than 11.0 gm/deciliter [13]. A child is classified to have undernutrition if they exhibit either stunting, wasting, underweight, or any of these conditions, as based on the composite index of anthropometric failure (CIAF) concept [27]. In the NDHS survey, weight was measured using SECA scales (model SECA874U). Height and length were measured with a measuring board (Shorrboard). Anemia was measured with blood specimens via finger-prick or heel-prick (for children 6-11 months of age). Hemoglobin was analyzed using a portable HemoCue device [13].

Children whose height-for-age z-score is below minus two standard deviations (-2 SD) from the median of the internal reference population were defined as stunted [24]. Children whose weight-for-height z-score is below -2 SD from the median of the reference population were defined as wasted [28]. Children whose weight-for-age z-score is below -2 SD from the median of the reference population were defined as underweight [28].

The outcome variable co-existence of undernutrition and anemia consists of four categories a) Normal (having neither undernutrition nor anemia) b) only undernutrition (having only undernutrition) c) only anemia (having only anemia) and d) co-existence of undernutrition-anemia (having both undernutrition and anemia).

Exposure variables

The exposure variables for this study were wealth quintile (poorest/poorer/middle/richer/richest), maternal body mass index (underweight/normal/overweight or obese), mother's education (no education/ basic level education/Secondary and higher level), mother's exposure to health programs in television and radio (yes/no), and mother's participation in household decision making (yes/no). The definition of each exposure variable is explained in table 1 below:

164 **Table 1: List of exposure variables with their definition**

Variables	Definition
Wealth quintile	The wealth quintile measures the economic status of the household, and it is classified into poorest, poorer, middle, richer, and richest based on the wealth index [13].
Mother's body mass index (BMI)	The mother's BMI was classified as underweight, normal, overweight, and obese based on the body mass index of the mother. The mother will be considered underweight if BMI is less than 18.5 kg/m ² , normal if BMI is between 18.5 to 24.9 kg/m ² , overweight if her BMI is between 25-29.9 kg/m ² and obese if her BMI is greater than 30.0 kg/m ² [13]
Mother's education	Mother's education was classified into basic education (grades 1 to 8), secondary level education (grades 9 to 12), or higher education (grades 13 and above) based on the DHS questionnaire [13].
Mother's exposure to Health program in TV/radio	The women were considered to have exposure to specific health programs if they heard or saw health programs on the radio or television (TV). This is measured based on the DHS questionnaire. These programs included eight different health-related programs broadcasted from radio and TV namely <i>Jana Swasthya Bahas</i> (Public Health Debate) television program, <i>Jeevan Chakra</i> (Life Cycle) television serial, <i>Jana Swasthya</i> (Public Health) radio program, <i>Swasthya Gatibidhi</i> (Health Affairs) radio program, <i>Eak Dui Tin Sunau eekai Chhin</i> (Listen for a While) radio program, <i>Bhanchhin Ama</i> (Mother Says) radio program, <i>Hello Bhanchin Ama</i> (Mother Says Hello) radio program and <i>Jeewan Rakshya</i> (Save Life COVID- response) radio program [13,29].
Mother's participation in household decision-making	A composite variable measured from the mother's participation (alone or with her husband) in making three household decisions (access to healthcare, major household purchases, and visits her family or relatives) grouped into no participation or participation in decision-making [13].
Anemia status in mother	Anemia was assessed using altitude-adjusted hemoglobin levels, and a mother was considered anemic if the level was less than 11.0 gm/deciliter. [13]

165

166 Potential Confounding variables

167 Based on the existing literature, some variables were found to be potential confounders in the
 168 association between outcome and exposure. Confounding variables included sociodemographic
 169 variables consisting of place of residence (rural/urban), and ecological belt (mountain/hill/terai), child-
 170 related variables consisting of the age of the child (in months) and sex of the child (male/female), and
 171 parents-related variables consisting of parity (Primipara/Multipara) and father's education (no
 172 education/basic level/secondary and higher level).

173

174 *Statistical analysis*

175 We conducted pre-analytical processing and statistical analysis using R version 4.3.2 [30] and R studio
 176 [31]. We carried out weighted descriptive and inferential analysis using the "survey" package to address
 177 complex survey design and non-response. We presented parametric numerical variables as mean
 178 (standard deviation) and non-parametric numerical variables as median (interquartile range) with their

179 95% confidence interval (CI). We presented categorical variables as frequency, percent, and their 95%
180 CI. We computed the prevalence and 95% CI using the Wilson method.

181 We performed multivariable multinomial logistic regression, to determine the association between the
182 co-existence of stunting, wasting, and anemia (normal/undernutrition only/ anemia only/ co-existence)
183 and predictor variables. We performed binary logistic regression to determine the association between
184 undernutrition or anemia with the predictor variables. We checked for multicollinearity using variance
185 inflation factor (VIF) and removed province (VIF= 13.31, greater than the cutoff of 10) from the
186 regression models, as the information provided by the province variable was contained by the region
187 and urban status variables. All regression analyses were conducted using complete cases after
188 excluding observations with missing data. We calculated and presented crude and adjusted odds ratio
189 (aOR) and their 95% CI.

Results

Table 2 presents the sociodemographic characteristics of 6-59 months of children stratified by age. Of the total U-5 children, 48.5% were female. Age distribution showed 10.4% were 6-12 months old, 45.8% were 1-3 years old, and 43.8% were 4-5 years old. Most children (59.1%) were from the Terai region, while the fewest (5.6%) were from the mountain region. The ratio of children from urban to rural areas was 2:1. By province, the majority were from Madhesh (26.4%), followed by Koshi (17.9%), Lumbini (16.7%), and Bagmati (15.4%).

Table 2: Characteristics of 6-59 months children

Characteristics	Overall n (%)	Late infant (6-12 months) n (%)	Toddler (1 to 3 years) n (%)	Pre-school age (4- 5 years) n (%)
<i>Unweighted N</i>	2395	256	1092	1047
<i>Weighted N</i>	2335	244	1069	1022
Age of children, n (%)		10.4	45.8	43.8
Sex				
Male	51.5	52.6	51.4	51.3
Female	48.5	47.4	48.6	48.7
Ecological region				
Mountain	5.6	5.6	5.6	5.6
Hill	35.3	41.7	34.4	34.7
Terai	59.1	52.7	60	59.8
Type of place of residence				
Urban	64.3	63.1	64.2	64.7
Rural	35.7	36.9	35.8	35.3
Province				
Koshi	17.9	18.7	17.2	18.5
Madhesh	26.4	20.7	27.2	27
Bagmati	15.4	20	15.8	13.9
Gandaki	7.2	6.7	6.1	8.5
Lumbini	16.7	15.6	17.1	16.6
Karnali	7.7	9.8	7.1	7.8
Sudurpashchim	8.6	8.5	9.6	7.7
Wealth quintile				
Poorest	24.5	22.2	24.4	25
Poorer	22	19.9	23.5	20.9
Middle	20.7	19.6	19.6	22
Richer	18.1	19.1	19.2	16.8
Richest	14.7	19.2	13.3	15.2
Mother education				
No education	22.2	15.3	23.1	22.9
Basic	35.9	32.5	35.7	37
Secondary and higher	41.9	52.2	41.2	40

<i>Missing</i>	55	0	14	40
Father's education				
No education	10.7	6.1	10.8	11.8
Basic	41	42.2	41.1	40.7
Secondary and higher	48.3	51.7	48.1	47.6
<i>Missing</i>	322	27	135	160
Mother's age at birth				
<20	19.8	18.3	19.2	20.8
20-34	76.3	78.9	77.7	74.2
35-49	3.9	2.8	3.1	5
<i>Missing</i>	259	21	104	134
Parity				
Primipara	32.1	41.3	37.3	24.2
Multipara	67.9	58.7	62.7	75.8
<i>Missing</i>	259	21	104	134
Mother's BMI				
Normal (18.5 to 24.9 kg/m ²)	61.7	58.1	62.2	62.1
Overweight or obese (≥ 25 kg/m ²)	23.7	29.7	20	26
underweight (< 18.5 kg/m ²)	14.6	12.2	17.8	11.9
<i>Missing</i>	399	22	185	192
Mother's participation in household decision making				
No participation	33.5	32.5	34.6	32.6
Participation	66.5	67.5	65.4	67.4

n: weighted frequency; %: weighted percentage

Figure 1 presents the prevalence of stunting, wasting and being underweight among Nepalese children aged 6-59 months. The prevalence of stunting, wasting, and underweight were 25.9% (95% CI: 23.6, 28.3), 7.8% (95% CI: 6.5, 9.3), and 19.8% (95% CI: 17.6, 22.1), respectively. Of the total children, 3.2% (95% CI: 2.5, 4.3) had all three conditions, 13.5% (95% CI: 11.9, 15.3) had two of the three conditions, and 16.8% (95% CI: 15.2, 18.5) had one of the three conditions.

[Figure 1 here]

Figure 1: Prevalence of coexistence of different forms of undernutrition among A) children aged 6-12 months, B) children aged 1-3 years, C) children aged 4-5 years and D) all children aged 6 months to 5 years.

The prevalence of anemia was 43.4% (95% CI: 40.9, 45.9), which included 18.8% (95% CI: 16.8, 20.9) with moderate to severe anemia and 24.6% (95%CI: 22.7, 26.6) with mild anemia. The prevalence of moderate to severe anemia was highest in late infants, accounting for 39.9% (95% CI: 33.2, 47.1) and decreased to 8.4% (95% CI: 6.7, 10.8) in preschools (**Figure 2**).

[Figure 2 here]

Figure 2: Prevalence of anemia among A) children aged 6-11 months, B) children aged 1-3 years, C) children aged 4-5 years and D) all children aged 6 months to 5 years.

The prevalence co-existence of stunting-anemia, wasting-anemia, underweight-anemia, undernutrition-anemia were 12.5% (95%CI: 10.9 to 14.3), 4.0% (95%CI: 2.9 to 4.9), 9.7% (95% CI: 8.7 to 11.7), and 16.0% (95% CI: 14.2 to 18.0) respectively (**Figure 3A-D**). Of total children, 1.9% (95%CI: 1.3 to 2.6) had all three undernutrition conditions and anemia, 6.4% (95%CI: 5.3 to 7.8) had two undernutrition conditions and anemia, and 7.6% (95%CI: 6.6 to 8.9) had one undernutrition condition and anemia. (*not presented in tables*).

[Figure 3 here]

Figure 3: Prevalence of co-existence of anemia with stunting, wasting, underweight, and overall undernutrition

Table 3 presents the prevalence of coexistence of anemia and undernutrition by age group and sex of child. The prevalence of coexistence of undernutrition and anemia is highest in children 6-12 months (overall: 16.2%, male: 17.4%, female: 14.9%) followed by 1-3 years old children (overall: 22.0%, male: 20.4%, female: 23.7%). Prevalence of only anemia was highest in 6-12 months (overall: 54.9%, male: 50.2%, female: 60.1%) and only undernutrition in 4-5 years children (overall: 23.5% male: 21.0%, female: 26.2%).

Table 3: Co-existence of anemia and undernutrition conditions (n=2,335)

Characteristic	Overall, % (95%CI)	6-12 months, % (95%CI)	1-3 years, % (95%CI)	4-5 years, % (95%CI)
All children				
None	39.1 (36.6, 41.7)	25.0 (19.4, 31.6)	33.6 (30.5, 36.9)	48.2 (44.3, 52.1)
Only Anemia	27.4 (25.3, 29.6)	54.9 (48.3, 61.3)	29.5 (26.4, 32.7)	18.6 (15.8, 21.8)
Only undernutrition	17.5 (15.6, 19.6)	3.9 (1.99, 7.49)	14.9 (12.5, 17.6)	23.5 (20.6, 26.8)
Co-existence pf undernutrition and anemia	16.0 (14.1, 18.0)	16.2 (11.7, 22.0)	22.0 (19.1, 25.3)	9.6 (7.89, 11.7)
Male children				
None	39.5 (36.0, 43.1)	29.5 (21.0, 39.6)	33.5 (29.0, 38.4)	48.2 (43.0, 53.5)
Only Anemia	27.9 (25.1, 30.9)	50.2 (41.1, 59.2)	30.7 (26.6, 35.1)	19.6 (15.9, 23.9)
Only undernutrition	16.5 (14.2, 19.2)	2.9 (0.94, 8.83)	15.4 (12.3, 19.2)	21.0 (17.2, 25.5)
Co-existence pf undernutrition and anemia	16.1 (13.7, 18.8)	17.4 (11.3, 25.9)	20.4 (16.9, 24.5)	11.2 (8.71, 14.2)
Female children				
None	38.7 (35.4, 42.1)	20.1 (13.3, 29.3)	33.8 (29.5, 38.4)	48.1 (43.0, 53.3)
Only Anemia	26.8 (23.8, 30.1)	60.1 (49.2, 70.0)	28.2 (23.9, 33.1)	17.7 (13.8, 22.3)
Only undernutrition	18.6 (15.9, 21.5)	4.9 (2.09, 11.2)	14.3 (11.2, 18.1)	26.2 (21.7, 31.2)
Co-existence pf undernutrition and anemia	15.9 (13.4, 18.7)	14.9 (9.01, 23.5)	23.7(19.3, 28.7)	8.0 (5.72, 11.1)

%; weighted percent; CI: confidence interval

Figure 4 and **Table 4** present the distribution of anemia, undernutrition and co-existence across categories of quintile wealth, mother's education, household decision making and mother's exposure to health programs on TV and radio. The prevalence of undernutrition and co-existence was highest in poorest wealth quintile and lowest in richest wealth quintile, and highest in children whose mother had no education.

[Figure 4 here]

Figure 4: Distribution of anemia, overall undernutrition and their co-existence of anemia across exposure variables (vertical bars represent 95% CIs)

Table 4: Distribution of undernutrition, anemia and their coexistence across exposure variables

Characteristics	6-12 months			1-3 year			4-5 years		
	Undernutrition only	Anemia only	Co-existence	Undernutrition	Anemia	Co-existence	Undernutrition	Anemia	Co-existence
Wealth quintile									
Poorest	55.9	4.3	19.0	24.2	19.7	26.6	12.3	30.9	13.2
Poorer	51.7	1.8	15.2	30.3	16.4	25.4	17.6	27.4	11.0
Middle	60.9	3.3	12.2	31.0	12.3	21.3	24.4	27.3	10.1
Richer	56.9	4.0	24.0	35.2	12.5	18.5	29.5	13.0	5.9
Richest	48.8	6.2	10.4	27.2	10.4	13.7	10.2	12.4	5.3
Mother education									
No education	51.7	8.1	20.9	27.5	15.1	35.5	21.9	29.0	15.6
Basic	51.1	1.1	22.4	27.7	16.1	22.9	17.7	27.9	9.4
Secondary and higher	58.1	4.4	11.0	32.3	13.8	13.9	18.0	15.9	7.2
Participation in household decision	52.9	4.7	14.9	28.5	14.9	21.3	17.9	22.2	9.4
Exposure to health programs on TV/radio	64.3	4.7	11.9	24.6	15.5	14.2	16.6	26.7	10.0
Mother's BMI									
Normal	58.0	2.9	17.1	30.3	14.4	21.1	15.7	26.2	10.8
Overweight or obese	46.1	5.0	13.9	37.7	10.2	12.9	22.1	18.8	5.5
Underweight	50.3	6.9	25.3	22.1	22.5	33.4	28.3	22.5	13.2

Table 5 presents the factors associated with undernutrition and anemia among children aged 6-59 months. In univariable regression, undernutrition was significantly associated with wealth quintile, mother's education, age of child, and maternal nutritional status. In multivariable regression, undernutrition was found to be associated with age of child, wealth quintile, mother's education, and maternal nutrition status after adjusting for confounding variables. The odds of having undernutrition is 1.96 times among 1-3 years child and 1.70 times in the 4-5 years children compared to 6-11 months children. The odds of undernutrition were 44% (aOR: 0.56, 95%CI: 0.34 to 0.92) lower in children from the richer wealth quintile compared to the poorest, 36% (aOR: 0.64, 95%CI: 0.45 to 0.93) lower in the child whose mother has secondary or higher-level education. We didn't find a significant association

between undernutrition and mother participation in household decisions and exposure to health programs on TV/radio.

Table 5: Factors associated with undernutrition among children aged 6-59 months

Characteristic	Presence of undernutrition, n (%) *	Unadjusted			Adjusted [#]		
		OR	95% CI	p-value	OR	95% CI	p-value
Age of child							
6-12 months	49 (20.1)	Ref			Ref		
1 year to 3 years	394 (36.9)	2.32	1.67, 3.22	<0.001	1.96	1.33, 2.90	<0.001
4 to 5 years	339 (33.2)	1.97	1.39, 2.81	<0.001	1.7	1.12, 2.59	0.013
Wealth quintile							
Poorest	246 (43.1)	Ref			Ref		
Poorer	195 (38.1)	0.81	0.62, 1.06	0.122	0.93	0.67, 1.30	0.680
Middle	162 (33.6)	0.67	0.50, 0.88	0.005	0.78	0.54, 1.13	0.194
Richer	109 (25.7)	0.46	0.32, 0.66	<0.001	0.53	0.34, 0.82	0.005
Richest	70 (20.2)	0.33	0.22, 0.50	<0.001	0.56	0.34, 0.92	0.024
Mother education							
No education	234 (46.3)	Ref			Ref		
Basic	301 (36.7)	0.67	0.52, 0.88	0.003	0.87	0.65, 1.16	0.332
Secondary and higher	231 (24.2)	0.37	0.28, 0.49	<0.001	0.64	0.45, 0.93	0.019
Mother's Participation in household decision		Ref			Ref		
No participation	279 (35.7)	0.86	0.70, 1.07	0.173	0.82	0.63, 1.06	0.133
Participation	503 (32.4)						
Exposure to health programs on TV/radio		Ref			Ref		
No	615 (34.5)	0.85	0.67, 1.09	0.199	0.92	0.69, 1.24	0.595
Yes	145 (31.0)						
Mother's BMI		0.57	0.43, 0.75	<0.001	0.75	0.56, 1.01	0.057
Normal	411 (34.5)	1.66	1.23, 2.23	<0.001	1.48	1.09, 2.01	0.013
Overweight or obese	106 (23.1)	Ref			Ref		
Underweight	132 (46.6)	2.32	1.67, 3.22	<0.001	1.96	1.33, 2.90	<0.001

n: weighted frequency; %: weighted percent; Ref: reference group; OR: odds ratio; aOR: adjusted odds ratio; CI: confidence interval; * Row percent

[#] adjusted for age, sex of child, father's education, parity, place of residence, ecological belt, mother's age at childbirth, and adjusted for each other

Table 6 presents the factors associated with anemia among children aged 6-59 months. The factors associated were age of the child, wealth quintile, and anemia status of the mother. The odds of having anemia were 41% (aOR: 0.59, 95%CI: 0.36 to 0.97) lower among richest quintile compared to poorest, 64% (aOR: 0.36, 95%CI: 0.24 to 0.53) lower among child aged 1-3 years and 87% (aOR: 0.13; 95%CI: 0.09 to 0.20) lower among 4-5 years children compared to 6-12 months children after adjusting for other independent variables. The odds of having anemia were 1.63 (95%CI: 1.27 to 2.10) times as many children whose mothers were anemic.

281 **Table 6: Factors associated with anemia among children aged 6-59 months**

Characteristic	Prevalence of anemia, n (%) *	Unadjusted			Adjusted#		
		OR	95% CI	p-value	OR	95% CI	p-value
Age of child							
6-11 months	173 (71.1)	Ref			Ref		
1 year to 3 years	551 (51.5)	0.43	0.31, 0.60	<0.001	0.36	0.24, 0.53	<0.001
4 to 5 years	289 (28.3)	0.16	0.11, 0.23	<0.001	0.13	0.09, 0.20	<0.001
Wealth quintile							
Poorest	239 (41.8)	Ref			Ref		
Poorer	233 (45.5)	1.16	0.88, 1.53	0.291	0.88	0.63, 1.23	0.451
Middle	222 (46.0)	1.19	0.90, 1.56	0.215	0.89	0.61, 1.30	0.531
Richer	209 (49.2)	1.35	1.02, 1.80	0.038	1.07	0.72, 1.60	0.725
Richest	110 (31.9)	0.65	0.46, 0.93	0.02	0.59	0.36, 0.97	0.037
Mother's education							
No education	265 (52.4)	Ref			Ref		
Basic	347 (42.4)	0.67	0.52, 0.86	0.002	0.80	0.57, 1.13	0.210
Secondary and Higher	388 (40.6)	0.62	0.48, 0.80	<0.001	0.84	0.56, 1.26	0.409
Mother's Participation in decision making							
No participation	365 (46.6)	Ref			Ref		
Participation	648 (41.7)	0.82	0.65, 1.03	0.095	0.76	0.55, 1.05	0.099
Exposure to health programs on TV/radio							
No	813 (45.6)	Ref			Ref		
Yes	179 (38.4)	0.74	0.59, 0.94	0.012	1	0.74, 1.35	0.990
Mother's BMI							
Normal	516 (43.2)	Ref			Ref		
Overweight or obese	189 (41.1)	0.92	0.70, 1.21	0.548	1.19	0.87, 1.63	0.286
Underweight	149 (52.5)	1.46	1.07, 1.98	0.016	1.20	0.84, 1.71	0.319
Anemia status in mother							
No	510 (38.8)	Ref			Ref		
Yes	406 (54.1)	1.86	1.49, 2.32	<0.001	1.63	1.27, 2.10	<0.001

n: weighted frequency; %: weighted percent

* Row percent

Ref: reference group; OR: odds ratio; AOR: adjusted odds ratio; CI: confidence interval

adjusted for age, sex of child, father's education, parity, place of residence, ecological belt, mother's age at childbirth and adjusted for each other

Table 7 presents the factors associated with co-existence of undernutrition and anemia among children aged 6-59 months. The co-existence of undernutrition and anemia is significantly associated with the richest wealth quintile, the mother's education status, the mother's BMI, and presence of anemia in children. The children from the richest wealth quintile were 53% (aOR: 0.49; 95%CI: 0.26 to 0.86) less likely to have co-existence of undernutrition and anemia than children from the poorest wealth quintile. The children whose mothers had secondary or higher-level education had 48% (aOR: 0.52; 95%CI: 0.32 to 0.93) lower odds of having co-existence compared to children whose mothers had no formal education, whose mother's participation in decision-making in the family was 34% (aOR: 0.66; 95%CI: 0.47 to 0.94) lower odds of having co-existence after adjusting for confounding variables. The child whose mother was classified as underweight (BMI<18 kg/m²) was 1.80 times (95%CI: 1.20 to 2.70)

more likely and mother with anemia were 2.43 times more likely to have co-existence compared to a child whose mother had normal nutrition status.

Table 7: Factors associated with co-occurrence of undernutrition and anemia among children aged 6-59 months

Exposure variables	Anemia only vs Normal			Undernutrition only vs Normal			Coexistence vs Normal		
	aOR [#]	95% CI	p-value	aOR [#]	95% CI	P-value	aOR [#]	95% CI	p-value
Age of child (ref: 6-11 months chi)									
1-3 years	0.37	0.25 – 0.54	<0.001	3.02	1.32 – 6.91	0.009	0.68	0.42 – 1.11	0.123
45 years	0.16	0.11 – 0.24	<0.001	3.16	1.39 – 7.20	0.006	0.21	0.12 – 0.35	<0.001
Wealth Quintile (ref: Poorest)									
Poorer	0.94	0.63 – 1.40	0.77	0.98	0.65 – 1.48	0.928	0.8	0.51 – 1.25	0.325
Middle	0.98	0.64 – 1.49	0.917	0.83	0.53 – 1.30	0.412	0.61	0.37 – 1.01	0.054
Richer	1.04	0.67 – 1.61	0.869	0.48	0.28 – 0.81	0.006	0.58	0.34 – 0.99	0.048
Richest	0.49	0.30 – 0.79	0.004	0.38	0.21 – 0.67	0.001	0.47	0.26 – 0.86	0.015
Mother's education (ref: no education)									
Basic	0.87	0.60 – 1.27	0.473	1.00	0.67 – 1.49	0.996	0.70	0.46 – 1.06	0.094
Secondary and higher	1.02	0.67 – 1.56	0.923	0.87	0.54 – 1.38	0.543	0.52	0.32 – 0.86	0.010
Participation in HH decision (ref: nonparticipation)	0.69	0.51 – 0.93	0.016	0.73	0.52 – 1.02	0.064	0.66	0.47 – 0.94	0.02
Exposure to health programs on TV/radio (ref: No)	1.04	0.77 – 1.42	0.783	1.00	0.70 – 1.42	0.996	0.89	0.59 – 1.34	0.572
Mother's BMI (ref: average weight)									
Overweight and obese	1.30	0.97 – 1.74	0.076	0.89	0.63 – 1.28	0.539	0.88	0.59 – 1.32	0.529
Underweight	1.17	0.80 – 1.71	0.426	1.44	0.96 – 2.16	0.075	1.80	1.20 – 2.70	0.005
Anemia in mother (ref: No)	1.55	1.19 – 2.02	0.001	1.41	1.04 – 1.89	0.025	2.43	1.78 – 3.32	<0.001

Normal: participants having neither undernutrition nor anemia

ref: reference group; **aOR:** adjusted odds ratio; **CI:** confidence interval

Bold represents significance at 95% confidence level

[#] adjusted for age, sex of child, father's education, parity, place of residence, ecological belt, mother's age at childbirth, and adjusted for each other

Discussion:

Our analysis of a national survey revealed significant issues in undernutrition, anemia and their co-existence among Nepalese children in 2022. Overall, the prevalence of undernutrition and anemia was 33.5% and 43.4% respectively. The prevalence of coexistence between undernutrition and anemia was 16.0%, and for the co-existence of stunting, wasting, and underweight with anemia was 12.5%, 4.0% and 9.7% respectively. Undernutrition demonstrated a significant association with household wealth, maternal education, parity, and mother's BMI. Anemia was associated with both wealth and mother's anemia status. The co-existence of undernutrition and anemia exhibited significant associations with wealth, maternal education, maternal involvement in household decision-making, parity, mother's BMI, and the child's anemia status. However, undernutrition, anemia, and their co-occurrence were not found to be associated with maternal exposure to health programs on television and radio.

A study in Ethiopia reported a higher prevalence of undernutrition among preschool children (50.8%) and children aged 6-59 months (57.3%) [32] reported higher prevalence of undernutrition compared to the findings from our study where we found prevalence of undernutrition among 6-59 months children in Nepal to be 33.5%. Our findings indicate that Nepal's burden, though lower, remains substantial. These differences may reflect variations in economic development, food security, and maternal and child health interventions across countries [33].

The prevalence of anemia among 6-59 months children from our study (43.5%) is higher than the global prevalence (39.8%) but lower than the prevalence in African children (60.2%)[12]. Our study also demonstrates a 9.1 percentage-point reduction in anemia prevalence between 2016 (52.6%) [34] and 2022, indicating progress in addressing this issue. Comparable rates have been observed in neighboring countries. A meta-analysis conducted by Kundu et al in Bangladesh showed a prevalence of 46.8% [35] which was similar to our finding. The latest national family health survey (2019-21) in India reported 67.1% prevalence of anemia [36] which is more than what we found in Nepal. The decline in anemia prevalence in Nepal [23] is encouraging but underscores the need for sustained efforts to address this condition, particularly among vulnerable populations.

A systematic review reported the pooled prevalence of wasting-anemia and stunting-anemia in least-developed countries to be 5.4% and 19.5%, respectively[37]. These rates are higher than our findings from Nepal, which were 4.0% and 12.5%, respectively.

The high prevalence of co-existence of undernutrition and anemia in our study (16%) suggests that affected children are trapped in a vicious cycle of micronutrients. Undernutrition impairs iron absorption and utilization, while anemia exacerbates undernutrition by reducing the body's ability to effectively utilize nutrients.

The children from the richest wealth quintile had lower odds of having co-existence of anemia and undernutrition compared to the poorest wealth quintile. This wealth-related equity gap in undernutrition among children aged 6-59 is attributed to the economic status of households which affects the capability for food insecurity and health care utilization whenever the child is ill [38–40].

The children from a family in which mothers participate in decision-making and mothers who had secondary or higher education were less likely to have co-existence of anemia and undernutrition and anemia alone compared to children without anemia and undernutrition. The health and nutrition of the child highly depend on the mother, being the primary caretaker of the child with awareness of the child's health. The analysis of DHS data from Nepal and India suggests that women's decision-making authority (maternal autonomy in household decision-making) improves the nutritional status of children and reduces child mortality, even after controlling for education and wealth[41].

Strengths and Limitations

This study has several strengths. First, it utilized complex survey analysis to address complex survey design and non-response rates. Second, the findings from this study are generalizable to children aged 6-59 months in Nepal and to children of the same age group in demographically and economically similar countries, as it used a nationally representative dataset.

This study is not free from limitations. One of the limitations is due to the missing data we could not utilize some of the important dietary factors like minimum dietary diversity score and birth weight of child. Second, due to the cross-sectional nature of the study, the directionality of association is not possible to determine. Third, the results from this study may not be generalizable to populations outside Nepal. Lastly the wide confidence intervals observed in the associations between children's nutritional status and anemia suggest limited statistical power, likely due to small sample sizes within some subgroups.

Conclusion

The prevalence of undernutrition, anemia, and their co-existence remains high among children aged 6-59 months in Nepal. The co-existence of anemia and undernutrition was less likely among children from highest wealth quintile, those whose mothers attained at least secondary level education, and participated in household decision-making. Conversely, co-existence was more prevalent among children whose mothers had poor maternal nutritional status. It is essential to implement holistic programs and targeted interventions to empower poor families, improve mother's education, strengthen their roles in family ultimately improve the children's health.

376 **Data and analysis code availability**

377 We used publicly available data from Nepal Demographic and health survey database upon request
378 from the official website of “The DHS program”. The dataset is available at:

379 https://dhsprogram.com/data/dataset/Nepal_Standard-DHS_2022.cfm?flag=1

380 The analysis code and data used for this analysis is available on GitHub:

381 <https://github.com/Adrisyaone/Undernutritiona-and-anemia>

382

383

384 **Conflict of interest**

385 The authors declare no conflicts of interest.

386

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389 sectors.

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520 **Legends of tables and figures**

521 **Tables**

522 Table 1: List of exposure variables with their definition

523 Table 2: Characteristics of 6-59 months children

524 Table 3: Co-existence of anemia and undernutrition conditions (n=2,335)

525 Table 4: Distribution of undernutrition, anemia and their coexistence across exposure variables

526 Table 5: Factors associated with undernutrition among children aged 6-59 months

527 Table 6: Factors associated with anemia among children aged 6-59 months

528 Table 7: Factors associated with co-occurrence of undernutrition and anemia among children aged 6-59
529 months

530

531 **Figures**

532 Figure 1: Prevalence of coexistence of different forms of undernutrition among A) children aged 6-12
533 months, B) children aged 1-3 years, C) children aged 4-5 years and D) all children aged 6 months to 5
534 years.

535 Figure 2: Prevalence of anemia among A) children aged 6-11 months, B) children aged 1-3 years, C)
536 children aged 4-5 years and D) all children aged 6 months to 5 years.

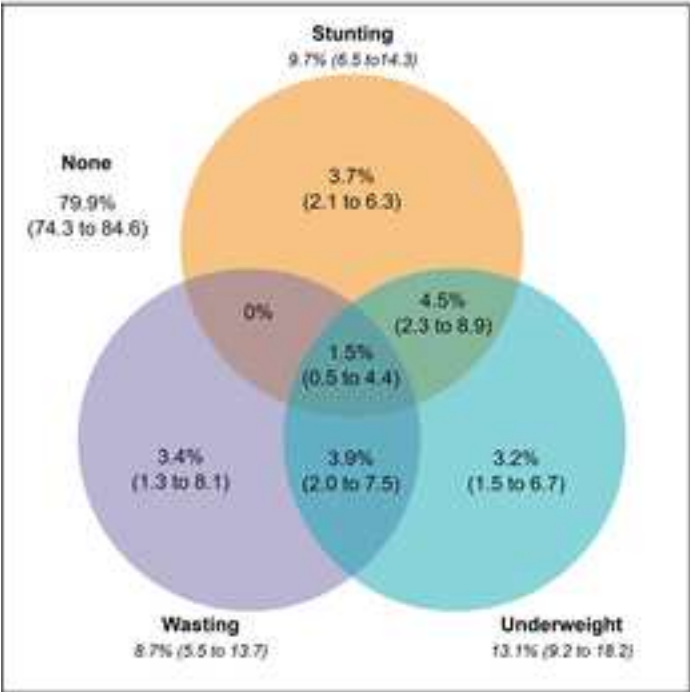
537 Figure 3: Prevalence of co-existence of anemia with stunting, wasting, underweight, and overall
538 undernutrition

539 Figure 4: Distribution of anemia, overall undernutrition and their co-existence of anemia across
540 exposure variables

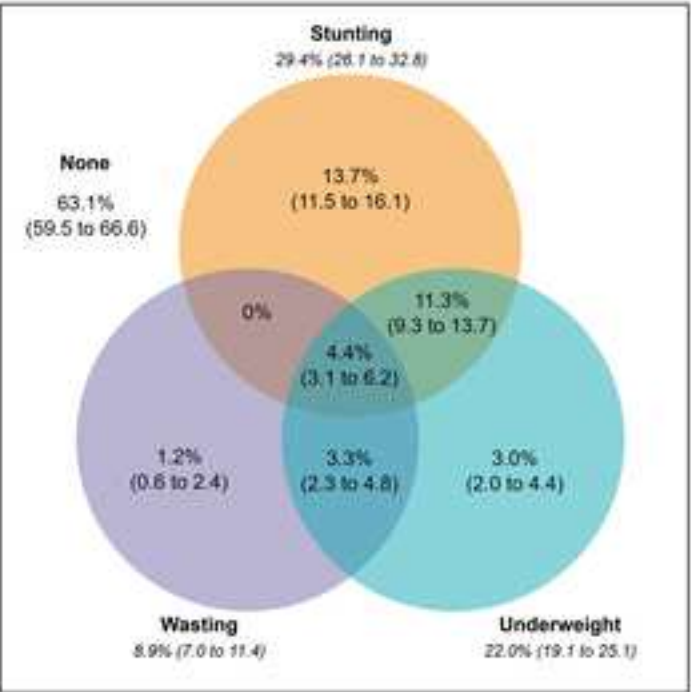
Figure 1: Prevalence of coexistence of different forms of undernutrition among A) children aged 6-12 months, B) children

[Click here to access/download;Figure;Figure 1.tif](#)

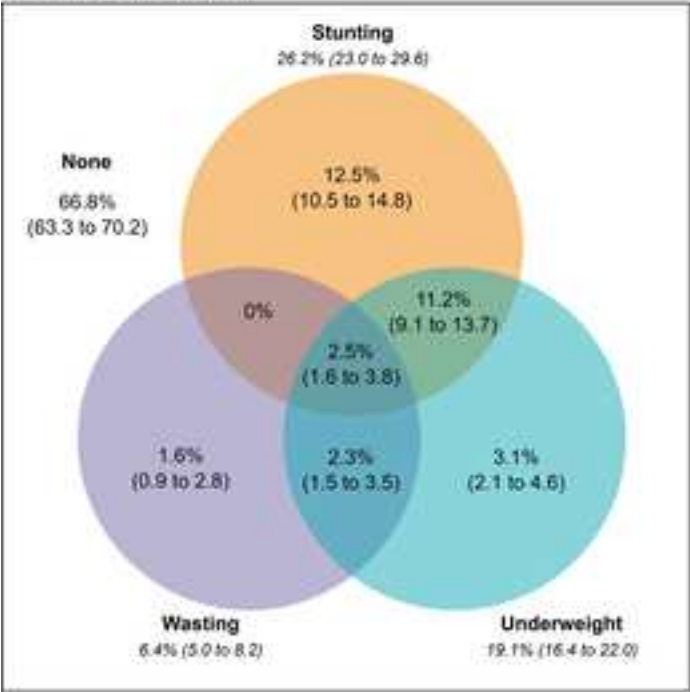
A. Children aged 6-12 months



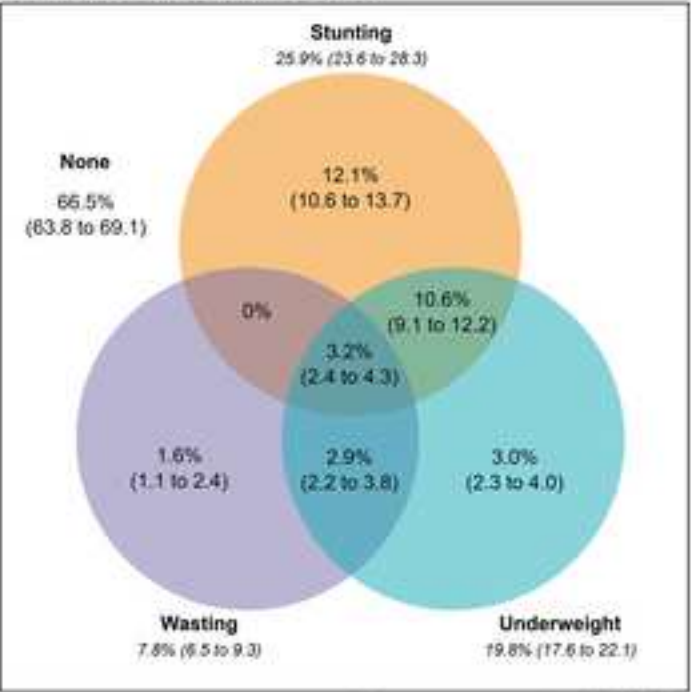
B. Children aged 1 to 3 years



C. Children aged 4-5 years



D. All children aged 6 months to 5 years

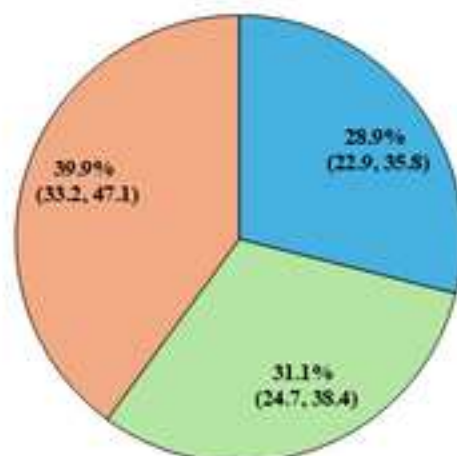


Values are expressed as % (95% CI)

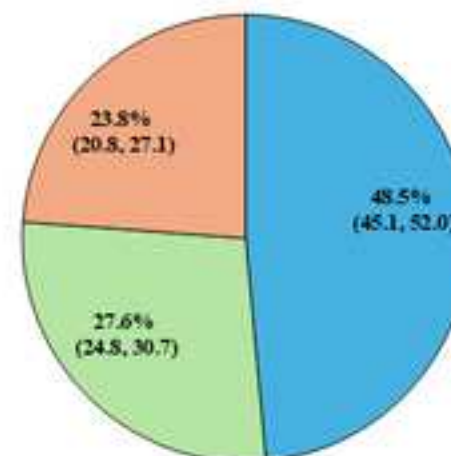
Figure 2: Prevalence of anemia among A) children aged 6-11 months, B) children aged 1-3 years, C) children aged 4-5 years and D) all children aged 6 months to 5 years.

[Click here to access/download;Figure;Figure 2.tif](#)

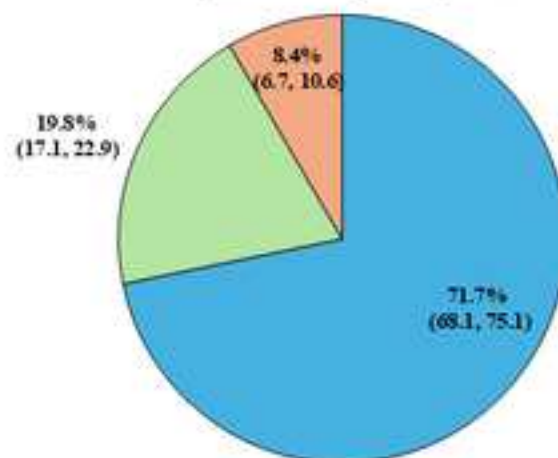
B. Children aged 6 months to 11 months (Late infants)



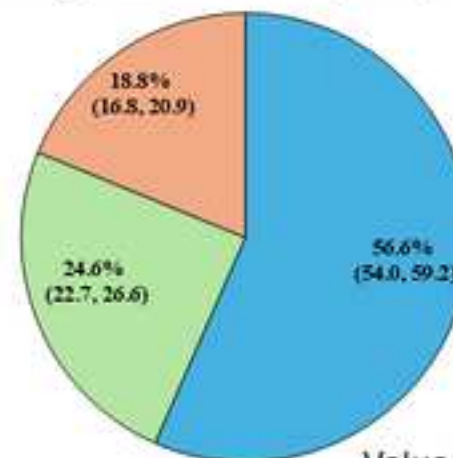
C. Children aged 1 year to 3 years (Toddlers)



D. Children aged 4 to 5 years (Preschool)



A. Children aged 6 months to 5 years (All)

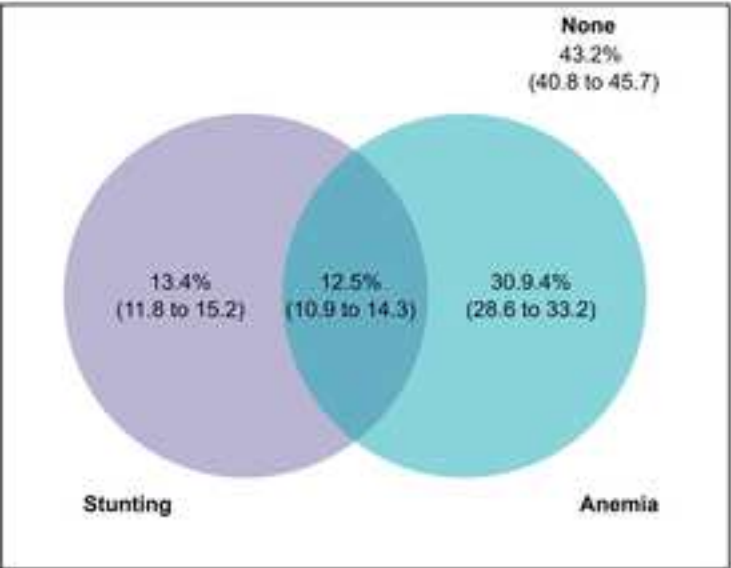


■ Normal
■ Mild
■ Moderate to severe

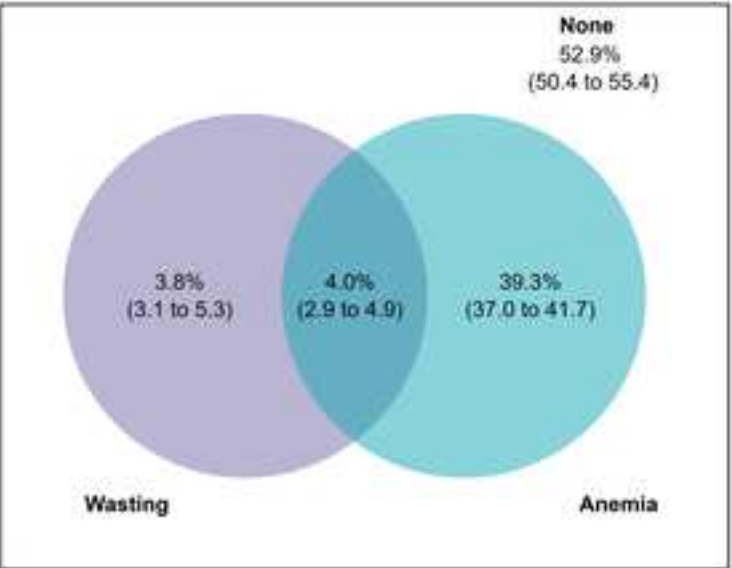
Values are expressed as % (95% CI)

Figure 3: Prevalence of co-existence of anemia with stunting, wasting, underweight, and overall undernutrition

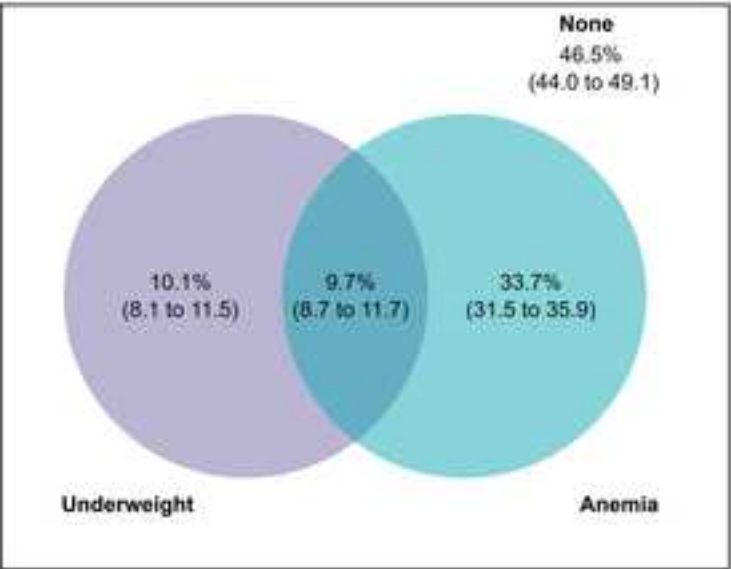
A. Co-existence of stunting and anemia



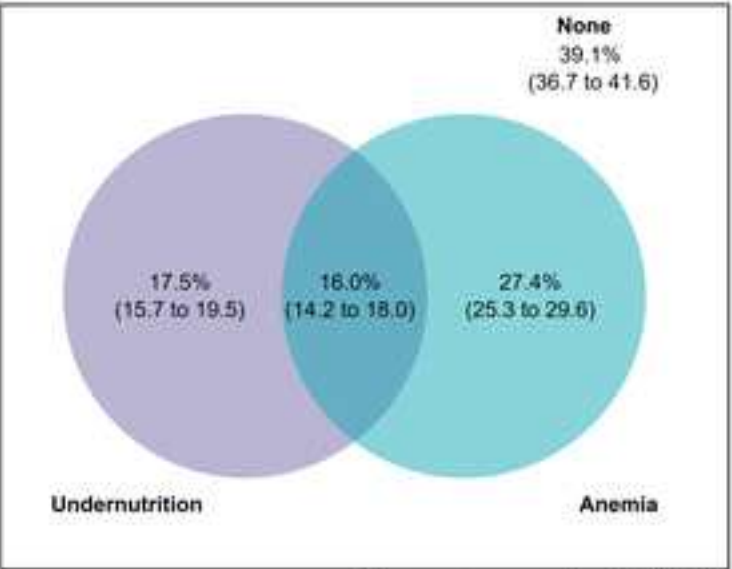
B. Co-existence of wasting and anemia



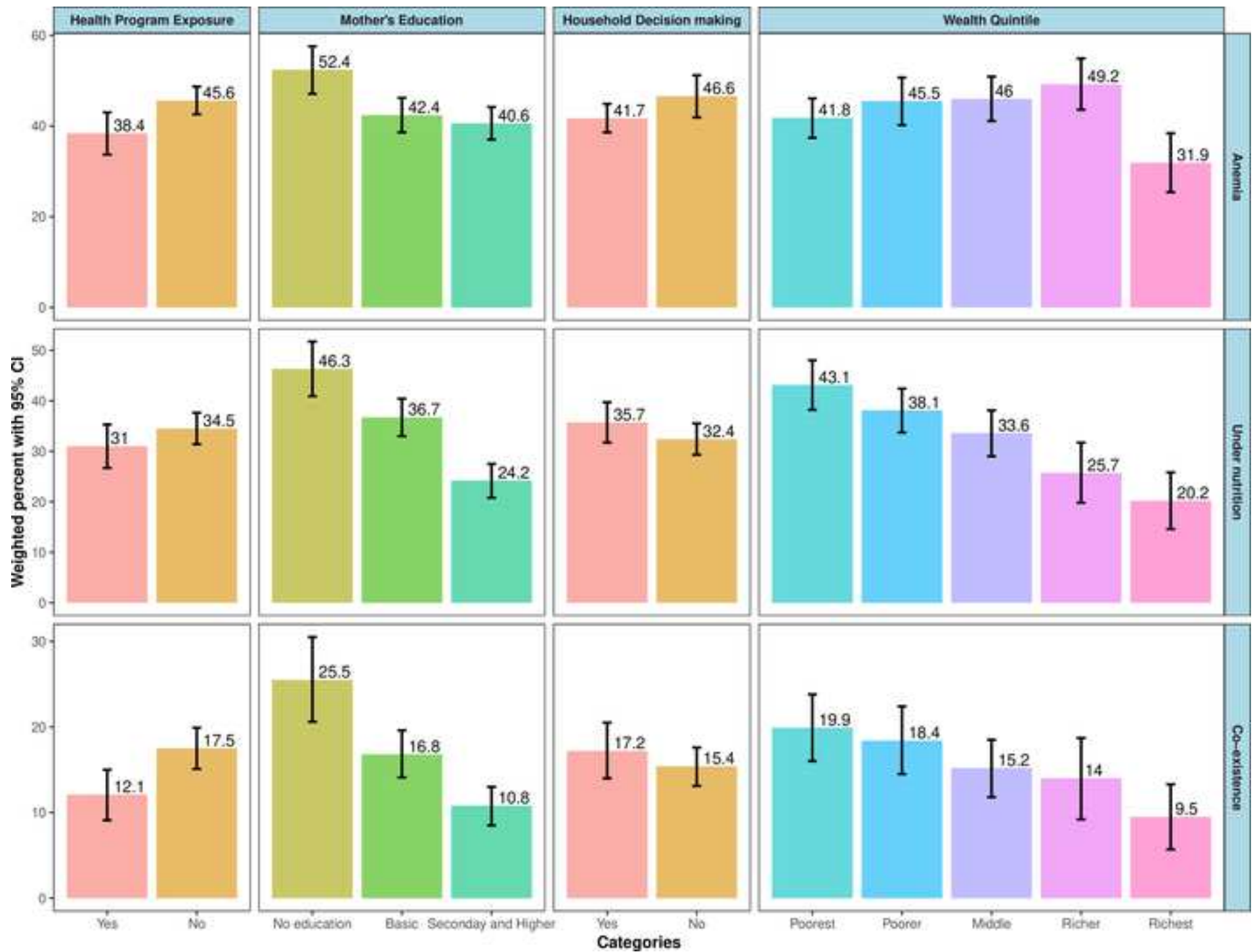
C. Co-existence of underweight and anemia



D. Co-existence of undernutrition and anemia



Values are expressed as % (95% CI)



1 ~~Factors Associated with~~**Determinants of Co-existence of Undernutrition and Anemia and**
2 ~~Their Co-existence among Children aged 6-59 Months~~**Children** in Nepal: ~~A~~
3 ~~Secondary Analysis of the Evidence from the 2022~~ Nepal Demographic and Health
4 Survey~~NDHS 2022~~

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21 **Word count:** ~~3886~~**5103**

22 **Number of tables:** ~~4~~**7**

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Abstract

Introduction: Undernutrition and anemia among children aged 6 to 59 months are significant public health issues in developing countries like Nepal. The co-existence of these conditions impacts growth and childhood development. This analysis study aimed to determine the prevalence of undernutrition, and anemia, and assess their co-existence, and identify in Nepal, as well as determine their contributing factors among children aged 6-59 months in Nepal.

Methods: We conducted a secondary data analysis using data from 2022 Nepal Demographic and Health Survey (NDHS 2022), a nationally representative of cross-sectional study survey performed a secondary analysis using analyzed data from the 2022 Nepal Demographic and Health Survey. The outcome variables were undernutrition, anemia, and their co-existence, of undernutrition with anemia. Children aged 6-59 months were considered undernourished if they exhibited either stunting, wasting, underweight or any combination of these conditions. Anemia was defined as hemoglobin levels <11.0 gm/dL (adjusted for altitude). We applied multivariable multinomial logistic regression to determine factors associated with co-existence, and multivariable logistic regression to assess factors associated with undernutrition and anemia separately. We presented the results from the regression analysis using adjusted odds-ratio or odds-like ratios (aOR) and 95% confidence intervals (CI).

Results: Among 233,523 children, the weighted prevalence of undernutrition was 33.5%, had undernutrition and of anemia was 43.4%, had anemia, and co-existence of 16.0% had undernutrition and anemia was co-existence 16.0%. Children from the richest wealth quintile, whose mothers had at least secondary education, and those whose mother participated in household decision-making had 53.4% (aOR: 0.479; 95%CI: 0.26 to 0.8694), 48.5% (aOR: 0.525; 95%CI: 0.323 to 0.93), and 34.7% (aOR: 0.663; 95%CI: 0.474 to 0.944) lower likelihood odds of experiencing co-existence of undernutrition and anemia compared to their counterparts. Children with malnourished mothers had 80.90% (aOR=1.80; 95% CI: 1.204 to 2.790) higher likelihood odds of co-existence of undernutrition and anemia compared to their counterparts. Anemia and undernutrition were not found to be associated with each other.

Conclusion: The prevalence of undernutrition, anemia, and their co-existence is among children was high in Nepal. Targeted interventions are essential to reduce these conditions and improve children's nutritional status.

Keywords: undernutrition; anemia; co-existence; 6-59-months children; Nepal; NDHS 2022

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59 Introduction

60 Undernutrition ~~encompassing including~~ stunting, wasting, ~~and~~ underweight ~~constitutes is~~ a major public
61 health challenge among children under 5 years (U-5) of age globally. This issue is particularly
62 pronounced in developing countries like Nepal [1–4]. ~~It results from~~ ~~There are~~ -three different causes –
63 immediate, underlying, and basic. ~~The~~ ~~Immediate~~ causes include inadequate diet intake and repeated
64 illnesses. ~~The~~ ~~The~~ underlying ~~factors~~ ~~causes~~ ~~causes~~ include suboptimal child feeding and caregiver
65 practices ~~consist of inadequate care~~ ~~fr~~om mothers, lack of access to healthcare facilities, poverty, and
66 food insecurity. ~~The~~ ~~The~~ basic causes include limited information, political and economic insecurity,
67 gender inequality, and the occurrence of natural disasters [5].

68 ~~Different indexes of undernutrition measure different underlying development status. Stunting, wasting,~~
69 ~~and being underweight are the indicators of poor nutritional status for children. While~~ ~~Stunting~~
70 indicates chronic malnutrition reflecting long-term nutrient deficiencies and is linked with delayed
71 motor development and impaired cognitive development [6]. ~~Wasting~~ indicates acute malnutrition
72 associated with recent severe food shortages or illnesses leading to weight loss and is a strong predictor
73 of mortality [6]. Underweight combines information about linear growth obstruction and weight for
74 length or height [7].

75 Anemia, particularly iron deficiency anemia, is another common and pressing public health concern
76 among children U-5 in low-, middle- and high-income countries. It results from poor nutrition and has
77 severe adverse health consequences including impaired cognitive development, impaired immunity,
78 disability, and increased risk of morbidity and mortality [8–10].

79 In 2022, the global prevalence of stunting among children under five years of age was 22.3%, while in
80 South Asia, it reached 31.8%. Additionally, the global prevalence of wasting was ~~reecorded~~ at 6.8%, with
81 South Asia reporting a prevalence of 14.8%. ~~[11]. In 2019, global anemia prevalence was 39.8%, which~~
82 ~~is equivalent to 269 million children aged 6–59 months~~ [12]. In Nepal, the prevalence of stunting, and
83 wasting among children under 5 was 25% and 8% respectively, and the prevalence of anemia was 43%
84 among children 6–59 months old [13].

85 The co-existence of undernutrition and anemia increases the risk of childhood morbidity and mortality
86 [14–16]. Almost fifty percent of deaths among children under the age of five are associated with
87 undernutrition, which predominantly occurs in low- and middle-income countries (LMICs). Notably,
88 88% of these countries (124 out of 141) experience multiple forms of malnutrition [17]. The
89 developmental, economic, social, and medical impacts of ~~the global burden of~~ undernutrition and
90 anemia among children are serious and ~~lasting~~ ~~perpetuating~~, for individuals and their families,
91 communities, and countries [7].

Furthermore, Anemia and undernutrition are both concentrated in socioeconomically disadvantaged groups, which and they share numerous multifaceted causes involving and complex interactions between diet, transmissible illnesses, and other factors, such as inadequate care and unhealthy household environments that adversely affect the cognitive development and physical well-being of children and may lead to increased mortality [18–20]. For instance, lower maternal education levels and socioeconomic status of households are linked with higher rates of malnutrition among children [18].

Though the concern on nutrition has advanced should be explored beyond the existence of any single form of malnutrition to the co-existence of multiple forms malnutrition [18], most studies in Nepal have of the studies focused on undernutrition and anemia separately. A few studies that have examined both conditions together have primarily reported that children from poorer households, mothers with low education levels, and those with inadequate dietary diversity are were more likely to experience concurrent undernutrition and anemia [21,22] [1, 31, (Dhakal, R.P., Aryal, V. & Acharya, P. Factors influencing undernutrition among children in Nepal: a comparative study of Composite Index of Anthropometric Failure (CIAF) using 2016 and 2022 demographic and health survey data. *J Health Popul Nutr* 44, 311 (2025). <https://doi.org/10.1186/s41043-025-00900-8>), (Guo W, Zgambo M, Chen S, Shimpuku Y. Prevalence and trends of coexisting forms of malnutrition and its associated factors among children aged 6–59 months in South and Southeast Asia, 1996–2022: a cross-sectional time series study. *BMC Public Health*. 2025 Jul 3;25(1):2296. doi: 10.1186/s12889-025-23482-w. PMID: 40610994; PMCID: PMC12224721)]. Even when both malnutrition and anemia were considered, anemia was often treated merely as a correlating factor or predictor rather than as part of a combined outcome reflecting their simultaneous occurrences. Moreover, there remains a gap in understanding how maternal factors (such as nutritional status and exposure to health information) and household-level socioeconomic characteristics interact to influence There are limited studies that explored the co-existence of undernutrition and anemia among U-5 children comprehensively. In addition, there are limited studies exploring factors associated with the co-existence of undernutrition and anemia among U-5 children under five in Nepal.

Therefore, This study aims to ascertain the prevalence of stunting, wasting, and underweight, as well as their co-occurrence, and the co-existence of undernutrition and anemia in Nepal. It also assesses the association of undernutrition, anemia, and co-existence of undernutrition with household economic status, maternal education and nutritional status, and exposure to televised or broadcasted health programs among children aged 6–59 months in Nepal. By identifying these factors, this study aims to help inform the development of targeted interventions and policies designed to mitigate malnutrition within this vulnerable population, thereby enhancing child health outcomes and alleviating the burden of malnutrition in Nepal.

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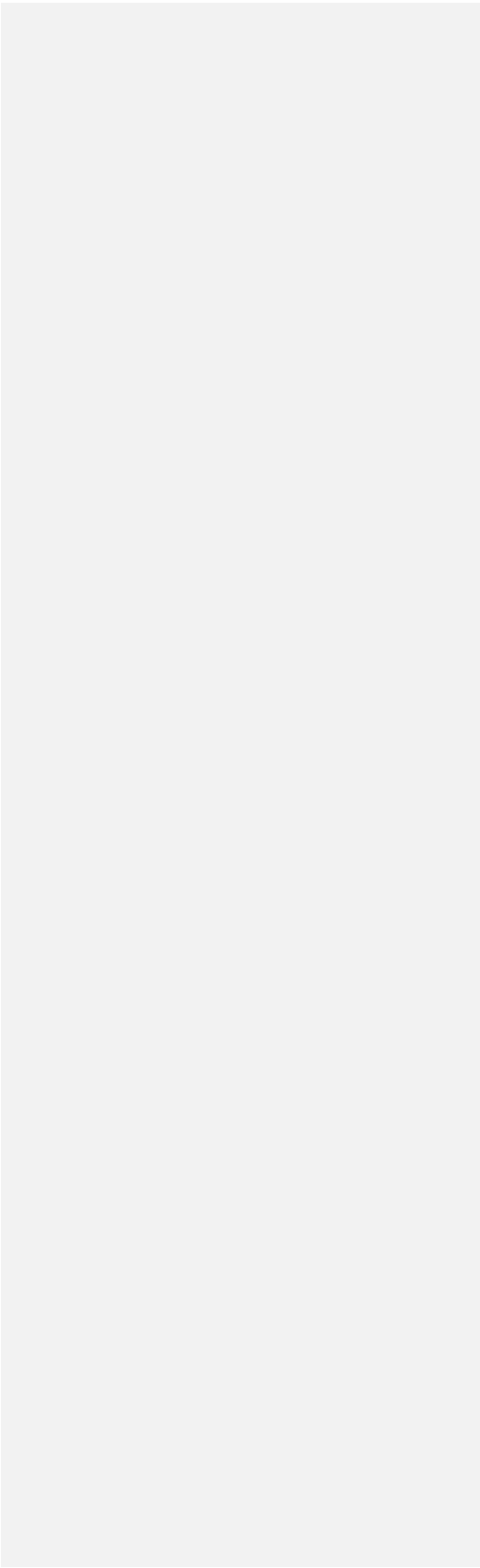
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128 **Methods**

129 *Data source*

130 In this study, we analyzed data from the Nepal Health Demographic Survey (NDHS) conducted in 2022
131 [23]. NDHS 2022 is the nationally representative survey implemented by New ERA under the aegis of
132 the Ministry of Health and Population (MoHP) with the technical support of ICF International and
133 funding from the United States Agency for International Development (USAID) [13]. This survey was
134 conducted to provide estimates of key demographic and health indicators at the national and provincial
135 levels. This study included 13,786 households, from which information on 14,845 women aged 15–49
136 years and 4,913 men aged 15–49 years and 2,643 children under 5 years.

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138 *Ethical approval*

139 We received permission from the official website of “the DHS program”
140 (<https://www.dhsprogram.com>) to download and use (submitted on: 07/25/2023 and approved on
141 07/26/2023) NDHS 2022 dataset [23]. NDHS 2022 obtained ethical approval from the institutional
142 review board of ICF International, United States of America (Reference number:
143 180657.0.001.NP.DHS.01, Date: 28th April 2022). and the ethical review board of Nepal Health
144 Research Council (Reference number: 678, Date: 30th September 2021) [13]. In the NDHS 2022,
145 informed consent was taken from the participants before enrolling them into the study.

147 *Study setting*

148 This study used nationally representative data from Nepal, a landlocked country located in Southeast
149 Asia with an area of 147, 516 km² [24]. It has seven administrative provinces, within which lies 753
150 municipalities (6 metropolitan cities, 11 sub-metropolitan cities, 276 urban municipalities, 460 rural
151 municipalities) [24]. Nepal has three ecological belts- Mountain, Hill, and Terai. Based on the 2021
152 Census, the total population of Nepal was 29,164,578 of which 14,911,027 (51.1 %) were females and
153 14,253,551 (48.9 %) were males [24]. Based on the Nepal Human Development Report 2020, Nepal
154 was ranked 145th based on the human development index (HDI) with an overall HDI of 0.587 and
155 of rural and urban parts of Nepal were of 0.647 and 0.561 respectively. with an overall HDI of 0.587
156 [25]. According to the Global Hunger Index, Nepal has a ranking of 19.5, implying moderate food
157 insecurity and undernourishment risk. [26].

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159 *Sample and sampling*

160 The NDHS 2022 used a two-stage stratified cluster sampling of households and stratification was
161 achieved based on rural and urban settings. In the first stage of sampling, Primary Sampling Units
162 (PSUs) were nominated by probability proportional to size followed by a systematic selection of
163 households from individual PSUs during the second stage of sampling. In this study, we will analyze
164 the data of 2,395 ~~children~~ children aged 6 months to 5 years ~~from the NDHS~~ using NDHS 2022 dataset.

166 *Measures*

167 Outcome variables

168 The primary outcome was “Coexistence of undernutrition (stunting, wasting or underweight) and
169 anemia”. ~~Anemia was assessed using altitude-adjusted hemoglobin levels and a child was considered~~
170 ~~anemic if the level was less than 11.0 gm/deciliter [13]. A child is classified to have undernutrition if~~
171 ~~they exhibit either stunting, wasting, underweight, or any of these conditions, as based on the composite~~
172 ~~index of a Anthropometric Failure (CIAF) concept [27]. In the NDHS survey, weight was measured~~
173 ~~using SECA scales (model SECA874U). Height and length were measured with a measuring board~~
174 ~~(Shorrboard â). Anemia was measured with blood specimens via finger-prick or heel-prick (for children~~
175 ~~6-11 months of age). Hemoglobin was analyzed using a portable HemoCueâ device [13].~~
176 ~~Children whose height-for-age z-score is below minus two standard deviations (-2 SD) from the median~~
177 ~~of the internal reference population were defined as stunted [-24]. Children whose weight-for-height z-~~
178 ~~score is below -2 SD from the median of the reference population were defined as wasted [28]. Children~~
179 ~~whose weight-for-age z-score is below -2 SD from the median of the reference population were defined~~
180 ~~as underweight [28].~~

182 The outcome variable co-existence of undernutrition and anemia consists of four categories a) Normal
183 (having neither undernutrition nor anemia) b) only undernutrition (having only undernutrition) c) only
184 anemia (having only anemia) and d) co-existence of undernutrition-anemia (having both undernutrition
185 and anemia).

186 ~~Children whose height for age z score is below minus two standard deviations (- 2 SD) from the median~~
187 ~~of the reference population were defined as stunted [24]. Children whose weight for height z score is~~
188 ~~below - 2 SD from the median of the reference population were defined as wasted [25]. Children whose~~
189 ~~weight for age z score is below -2 SD from the median of the reference population were defined as~~
190 ~~underweight[25].~~

191 ~~A child is classified as experiencing undernutrition if they exhibit stunting, wasting, underweight, or~~
192 ~~any combination of these conditions.[25] Anemia was assessed using altitude adjusted hemoglobin~~
193 ~~levels and a child was considered anemic if the level was less than 11.0 gm/deciliter [13].~~

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The outcome variable co-existence of undernutrition and anemia consists of four categories a) Normal (having neither undernutrition nor anemia) b) only undernutrition (having only undernutrition) c) only anemia (having only anemia) and d) co-existence of undernutrition anemia (having both undernutrition and anemia).

Exposure variables

The exposure variables for this study were wealth quintile (poorest/poorer/middle/richer/richest), mother's nutritional status/maternal body mass index (~~thin~~underweight/normal/overweight or obese), mother's education (no education/ basic level education/Secondary and higher level), mother's exposure to health programs in television and radio (yes/no), and mother's participation in household decision making (yes/no). The definition of each exposure variable is explained in table 1 below:

Table 1: List of exposure variables with their definition

Table 1: List of exposure variables with their definition

Variables	Definition
Wealth quintile	The wealth quintile measures the economic status of the household, and it is classified into poorest, poorer, middle, richer, and richest based on the wealth index [13].
Mother's body mass index (BMI) Mother's nutritional status	The mother's nutrition mother's BMI status was classified as thin underweight, normal, overweight, and obese based on the body mass index of the mother. The mother will be considered thin underweight if BMI is less than 18.5 kg/m ² , normal if BMI is between 18.5 to 24.9 kg/m ² , overweight if her BMI is between 25-29.9 kg/m ² and obese if her BMI is greater than 30.0 kg/m ² [13]
Mother's education	Mother's education was classified into basic education (grades 1 to 8), secondary level education (grades 9 to 12), or higher education (grades 13 and above) based on the DHS questionnaire [13].
Mother's exposure to Health program in TV/radio	The women were considered to have exposure to specific health programs if they heard or saw health programs on the radio or television (TV). This is measured based on the DHS questionnaire. These programs included eight different health-related programs broadcasted from radio and TV namely <i>Jana Swasthya Bahas</i> (Public Health Debate) television program, <i>Jeevan Chakra</i> (Life Cycle) television serial, <i>Jana Swasthya</i> (Public Health) radio program, <i>Swasthya Gatibidhi</i> (Health Affairs) radio program, <i>Eak Dui Tin Sunau eekai Chhin</i> (Listen for a While) radio program, <i>Bhanchhin Ama</i> (Mother Says) radio program, <i>Hello Bhanchin Ama</i> (Mother Says Hello) radio program and <i>Jeewan Rakshya</i> (Save Life COVID- response) radio program [13,29].
Mother's participation in household decision-making	A composite variable measured from the mother's participation (alone or with her husband) in making three household decisions (access to healthcare, major household purchases, and visits her family or relatives) grouped into no participation or participation in decision-making [13].
Anemia status in mother	Anemia was assessed using altitude-adjusted hemoglobin levels, and a mother was considered anemic if the level was less than 11.0 gm/deciliter. [13]

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Potential Confounding variables

Based on the existing literature, some variables were found to be ~~the~~ potential confounders in the association between outcome and exposure. Confounding variables included ~~They were~~ sociodemographic variables consisting of place of residence (rural/urban), and ecological belt (mountain/hill/terai), child-related variables consisting of the age of the child (in months) and sex of the child (male/female), and parents-related variables consisting of parity (Primipara/Multipara) and father's education (no education/basic level/secondary and higher level).

Statistical analysis

We conducted pre-analytical processing and statistical analysis using R version 4.3.2 [30] and R studio [31]. We carried out weighted descriptive and inferential analysis using the "survey" package to address complex survey design and non-response. We presented parametric numerical variables as mean (standard deviation) and non-parametric numerical variables as median (interquartile range) with their 95% confidence interval (CI). We presented categorical variables as frequency, percent, and their 95% CI. We computed the prevalence and 95% CI using the Wilson method.

~~All regression analyses were conducted using complete cases after excluding observations with missing data.~~ We performed multivariable multinomial logistic regression, to determine the association between the co-existence of stunting, wasting, and anemia (normal/undernutrition only/ anemia only/ co-existence) and predictor variables. We performed binary logistic regression to determine the association between undernutrition or anemia with the predictor variables. We checked for multicollinearity using variance inflation factor (VIF) and removed province ~~variable~~ (VIF= 13.31, greater than VIF> the cutoff of 10²) from the regression models, as the information provided by the province variable was contained by the region and urban status variables. ~~All regression analyses were conducted using complete cases after excluding observations with missing data.~~ We calculated and presented crude and adjusted odds ratio (aOR) ~~or odds like ratio~~ and their 95% CI.

Normal (18.5 to 24.9 kg/m ²)	61.74194 (61.7)	58.1	62.2	62.1
Overweight or obese- (>=25 kg/m ²)	23.7459 (23.7)	29.7	20	26
Thin-underweight (<18.5 kg/m ²)	14.6284 (14.6)	12.2	17.8	11.9
Missing	399399	22	185	192
Mother's participation in household decision making				
No participation	33.5783 (33.5)	32.5	34.6	32.6
Participation	66.51552 (66.5)	67.5	65.4	67.4

n: weighted frequency; %: weighted percentage

Figure 1A presents the prevalence of stunting, wasting and underweight among Nepalese children aged 6-59 months. The prevalence of stunting, wasting, and underweight were 25.9% (95% CI: 23.6, 28.3), 7.8% (95% CI: 6.5, 9.3), and 19.8% (95% CI: 17.6, 22.1), respectively. Of the total children, 3.2% (95% CI: 2.5, 4.3) had all three conditions, 13.5% (95% CI: 11.9, 15.3) had two of the three conditions, and 16.8% (95% CI: 15.2, 18.5) had one of the three conditions. The prevalence of anemia was 43.4% (95% CI: 40.9, 45.9), which included 18.8% with moderate to severe anemia and 24.6% with mild anemia (Figure 1B).

[Figure 1 here]

Figure 1: Prevalence of coexistence of different forms of undernutrition among A) children aged 6-12 months, B) children aged 1-3 years, C) children aged 4-5 years and D) all children aged 6 months to 5 years.

1: Preva

The prevalence of anemia was 43.4% (95% CI: 40.9, 45.9), which included 18.8% (95% CI: 16.8, 20.9 with) with moderate to severe anemia and 24.6% (95% CI: 22.7, 26.6) with mild anemia (Figure 1B). The prevalence of moderate to severe anemia was highest in late infants, accounting for 39.9% (95% CI: 33.2, 47.1) and decreased to 8.4% (95% CI: 6.7, 10.8) in preschools (Figure 2).

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[Figure 2 here]

Figure 2: Prevalence of anemia among A) children aged 6-11 months, B) children aged 1-3 years, C) children aged 4-5 years and D) all children aged 6 months to 5 years.

The prevalence co-existence of stunting anemia, wasting anemia, underweight anemia, undernutrition anemia were 12.5% (95%CI: 10.9 to 14.3), 4.0% (95%CI: 2.9 to 4.9), 9.7% (95% CI: 8.7 to 11.7), and 16.0% (95% CI: 14.2 to 18.0) respectively (*Figure 2A-D*). Of total children, 1.9% (95%CI: 1.3 to 2.6) had all three undernutrition conditions and anemia, 6.4% (95%CI: 5.3 to 7.8) had two undernutrition conditions and anemia, and 7.6% (95%CI: 6.6 to 8.9) had one undernutrition condition and anemia (*Table 2*).

Figure 2: Preva

The prevalence co-existence of stunting-anemia, wasting-anemia, underweight-anemia, undernutrition-anemia were 12.5% (95%CI: 10.9 to 14.3), 4.0% (95%CI: 2.9 to 4.9), 9.7% (95% CI: 8.7 to 11.7), and 16.0% (95% CI: 14.2 to 18.0) respectively (*Figure 3A-D*). Of total children, 1.9% (95%CI: 1.3 to 2.6) had all three undernutrition conditions and anemia, 6.4% (95%CI: 5.3 to 7.8) had two undernutrition conditions and anemia, and 7.6% (95%CI: 6.6 to 8.9) had one undernutrition condition and anemia (*not presented in tables*).

[Figure 3 here]

Figure 3: Prevalence of co-existence of anemia with stunting, wasting, underweight, and overall undernutrition.

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Table 3 presents the prevalence of coexistence of anemia and undernutrition by age group and sex of child. The prevalence of coexistence of undernutrition and anemia is highest in children 6-12 months (overall: 16.2%, male: 17.4%, female: 14.9%) followed by 1-3 years old children (overall: 22.0%, male: 20.4%, female: 23.7%). Prevalence of only anemia was highest in 6-12 months (overall: 54.9%, male: 50.2%, female: 60.1%) and only undernutrition in 4-5 years children (overall: 23.5% male: 21.0%, female: 26.2%).

{Figure 2 here}

Table 3: Co-existence of anemia and undernutrition conditions (n=2,335)

Characteristic	Overall, % (95%CI)	6-12 months, % (95%CI)	1-3 years, % (95%CI)	4-5 years, % (95%CI)
All children				
None	39.1 (36.6, 41.7)	25.0 (19.4, 31.6)	33.6 (30.5, 36.9)	48.2 (44.3, 52.1)
Only Anemia	27.4 (25.3, 29.6)	54.9 (48.3, 61.3)	29.5 (26.4, 32.7)	18.6 (15.8, 21.8)
Only undernutrition	17.5 (15.6, 19.6)	3.9 (1.99, 7.49)	14.9 (12.5, 17.6)	23.5 (20.6, 26.8)
Co-existence pf undernutrition and anemia	16.0 (14.1, 18.0)	16.2 (11.7, 22.0)	22.0 (19.1, 25.3)	9.6 (7.89, 11.7)
Male children				
None	39.5 (36.0, 43.1)	29.5 (21.0, 39.6)	33.5 (29.0, 38.4)	48.2 (43.0, 53.5)
Only Anemia	27.9 (25.1, 30.9)	50.2 (41.1, 59.2)	30.7 (26.6, 35.1)	19.6 (15.9, 23.9)
Only undernutrition	16.5 (14.2, 19.2)	2.9 (0.94, 8.83)	15.4 (12.3, 19.2)	21.0 (17.2, 25.5)
Co-existence pf undernutrition and anemia	16.1 (13.7, 18.8)	17.4 (11.3, 25.9)	20.4 (16.9, 24.5)	11.2 (8.71, 14.2)
Female children				
None	38.7 (35.4, 42.1)	20.1 (13.3, 29.3)	33.8 (29.5, 38.4)	48.1 (43.0, 53.3)
Only Anemia	26.8 (23.8, 30.1)	60.1 (49.2, 70.0)	28.2 (23.9, 33.1)	17.7 (13.8, 22.3)
Only undernutrition	18.6 (15.9, 21.5)	4.9 (2.09, 11.2)	14.3 (11.2, 18.1)	26.2 (21.7, 31.2)
Co-existence pf undernutrition and anemia	15.9 (13.4, 18.7)	14.9 (9.01, 23.5)	23.7 (19.3, 28.7)	8.0 (5.72, 11.1)

%. weighted percent; CI: confidence interval

Figure 4 and Table 4 present the distribution of anemia, undernutrition and co-existence across categories of quintile wealth, mother's education, household decision making and mother's exposure to health programs on TV and radio. The prevalence of undernutrition and co-existence was highest in poorest wealth quintile and lowest in richest wealth quintile, and highest in children whose mother had no education.

[Figure 4 here]

Figure 4: Distribution of anemia, overall undernutrition and their co-existence of anemia across exposure variables (vertical bars represent 95% CIs)

Table 4: Distribution of undernutrition, anemia and their coexistence across exposure variables

Characteristics	6-12 months			1-3 years			4-5 years		
	Undernutrition only	Anemia only	Co-existence	Undernutrition	Anemia	Co-existence	Undernutrition	Anemia	Co-existence
Wealth quintile									
Poorest	55.9	4.3	19.0	24.2	19.7	26.6	12.3	30.9	13.2
Poorer	51.7	1.8	15.2	30.3	16.4	25.4	17.6	27.4	11.0
Middle	60.9	3.3	12.2	31.0	12.3	21.3	24.4	27.3	10.1
Richer	56.9	4.0	24.0	35.2	12.5	18.5	29.5	13.0	5.9
Richest	48.8	6.2	10.4	27.2	10.4	13.7	10.2	12.4	5.3
Mother education									
No education	51.7	8.1	20.9	27.5	15.1	35.5	21.9	29.0	15.6
Basic	51.1	1.1	22.4	27.7	16.1	22.9	17.7	27.9	9.4
Secondary and higher	58.1	4.4	11.0	32.3	13.8	13.9	18.0	15.9	7.2
Participation in household decision	52.9	4.7	14.9	28.5	14.9	21.3	17.9	22.2	9.4
Exposure to health programs on TV/radio	64.3	4.7	11.9	24.6	15.5	14.2	16.6	26.7	10.0
Mother's BMI									
Normal	58.0	2.9	17.1	30.3	14.4	21.1	15.7	26.2	10.8
Overweight or obese	46.1	5.0	13.9	37.7	10.2	12.9	22.1	18.8	5.5
Underweight	50.3	6.9	25.3	22.1	22.5	33.4	28.3	22.5	13.2

Figure 3 presents the distribution of anemia, undernutrition and co-existence across categories of wealth quintile, mother's education, household decision making and mother's exposure to health programs on TV and radio. The prevalence of undernutrition and co-existence was highest in poorest wealth quintile and lowest in richest wealth quintile, and highest in children whose mother has no education.

Figure 3: Dis

[Figure 3 here]

Table 3-5 presents the factors associated with undernutrition and anemia among children aged 6-59 months. In univariable regression, undernutrition was significantly associated with wealth quintile,

mother's education, age of child, ~~place of residence, father's education, ecological belt, parity, and~~
maternal nutritional status, ~~and anemia in the child~~. In multivariable regression, undernutrition was
found to be associated with age of child, wealth quintile, mother's education, and maternal nutrition
status after adjusting for confounding variables. The odds of having undernutrition is 1.96 times among
1-3 years child and 1.70 times in the 4-5 years children compared to 6-11 months children. The odds of
undernutrition were 443% (aAOR: 0.567, 95%CI: 0.346 to 0.9289) lower in children from the richer
wealth quintile compared to the poorest, 365% (aAOR: 0.645, 95%CI: 0.45 to 0.934) lower in the child
whose mother has secondary or higher-level education. We didn't find a significant association between
undernutrition and mother participation in household decisions and exposure to health programs on
televisionTV/radio.

Table -5: Factors associated with undernutrition among children aged 6-59 months

Characteristic	Presence of undernutrition, n (%) *	Unadjusted			Adjusted [#]		
		OR	95% CI	p-value	OR	95% CI	p-value
Age of child							
6-12 months	49 (20.1)	Ref			Ref		
1 year to 3 years	394 (36.9)	2.32	1.67, 3.22	<0.001	1.96	1.33, 2.90	<0.001
4 to 5 years	339 (33.2)	1.97	1.39, 2.81	<0.001	1.7	1.12, 2.59	<0.013
Wealth quintile							
Poorest	246 (43.1)	Ref			Ref		
Poorer	195 (38.1)	0.81	0.62, 1.06	0.122	0.93	0.67, 1.30	0.680
Middle	162 (33.6)	0.67	0.50, 0.88	0.005	0.78	0.54, 1.13	0.190
Richer	109 (25.7)	0.46	0.32, 0.66	<0.001	0.53	0.34, 0.82	<0.001
Richest	70 (20.2)	0.33	0.22, 0.50	<0.001	0.56	0.34, 0.92	<0.001
Mother education							
No education	234 (46.3)	Ref			Ref		
Basic	301 (36.7)	0.67	0.52, 0.88	0.003	0.87	0.65, 1.16	0.272
Secondary and higher	231 (24.2)	0.37	0.28, 0.49	<0.001	0.64	0.45, 0.93	0.022
Missing	16						
Mother's Participation in household decision							
No participation	279 (35.7)	0.86	0.70, 1.07	0.173	0.82	0.63, 1.06	0.133
Participation	503 (32.4)	0.86	0.70, 1.07	0.173	0.83	0.62, 1.10	0.190
Exposure to health program on TV/radio							
No exposure	615 (34.5)	0.85	0.67, 1.09	0.199	0.92	0.69, 1.24	0.595

Yes	145 (31.0)	0.85	0.67 to 1.09	0.199	1.00	0.74 to 1.35	0.98
Missing	23	Ref			Ref		
Mother's BMI		0.57	0.43, 0.75	<0.001	0.75	0.56, 1.01	0.652
Maternal nutrition status							
Normal	411 (34.5)	1.66	1.23, 2.23	<0.001	1.48	1.09, 2.01	0.013
Overweight or obese	106 (23.1)	Ref	0.43 to 0.75	<0.001	Ref	0.56 to 1.02	0.066
Thin	132 (46.6)	2.32	1.67, 3.22	<0.001	1.96	1.33, 2.90	0.009
Underweight		1.66	1.23, 2.23	<0.001	1.48	1.09, 2.01	0.013
Missing	133	-	-	-	-	-	-
Anemia in child							
No	409 (30.9)	Ref			Ref		
Yes	373 (36.9)	1.3	1.07 to 1.59	0.009	1.19	0.92 to 1.54	0.184

n: weighted frequency; %: weighted percent; Ref: reference group; OR: odds ratio; aAOR: adjusted odds ratio; CI: confidence interval; * Row percent

adjusted for age, sex of child, father's education, parity, place of residence, ecological belt, mother's age at childbirth, and adjusted for each other

Table 4-6 presents the factors associated with anemia among children aged 6-59 months. The factors associated were age of the child, wealth quintile, age of the child, and anemia status of the mother. The odds of having anemia were 41% (aAOR: 0.597, 95%CI: 0.365 to 0.974) lower among richest quintile compared to poorest, 64% (aAOR: 0.364, 95%CI: 0.234 to 0.532) lower among child aged 1-3 years and 87% (aAOR: 0.13; 95%CI: 0.09 to 0.20) lower among 4-5 years children compared to 6-12 months children after adjusting for other independent variables. The odds of having anemia were 1.634 (95%CI: 1.267 to 2.110) times as many children whose mothers were anemic.

Table 6: Factors associated with anemia among children aged 6-59 months

Characteristic	Prevalence of anemia, n (%) *	Unadjusted			Adjusted#		
		OR	95% CI	p-value	OR	95% CI	p-value
Age of child							
6-11 months	173 (71.1)	Ref			Ref		
1 year to 3 years	551 (51.5)	0.43	0.31, 0.60	<0.001	0.36	0.24, 0.53	<0.001
4 to 5 years	289 (28.3)	0.16	0.11, 0.23	<0.001	0.13	0.09, 0.20	<0.001
Wealth quintile							
Poorest	239 (41.8)	Ref			Ref		
Poorer	233 (45.5)	1.16	0.88, 1.53	0.29	0.88	0.63, 1.23	0.52
Middle	222 (46.0)	1.19	0.90, 1.56	0.21	0.89	0.61, 1.30	0.54
Richer	209 (49.2)	1.35	1.02, 1.80	0.038	1.07	0.72, 1.60	0.98
Richest	110 (31.9)	0.65	0.46, 0.93	0.02	0.59	0.36, 0.97	0.02
Mother's education							
No education	265 (52.4)	Ref			Ref		

356

357 Table 5-7 presents the factors associated with co-existence of undernutrition and anemia among children
358 aged 6-59 months. The co-existence of undernutrition and anemia is significantly associated with the
359 richest wealth quintile, the mother's education status, the mother's ~~nutritional status~~ BMI, and presence
360 of anemia in children. The children from the richest wealth quintile were ~~51.53%~~ (aOR: 0.49; 95%CI:
361 0.26 to 0.~~8694~~) less likely to have co-existence of undernutrition and anemia than children from the
362 poorest wealth quintile. The children whose mothers had ~~ve~~ secondary or higher-level education had ~~ve~~
363 ~~48.45%~~ (aOR: 0.525; 95%CI: 0.323 to 0.93) lower ~~oddslikelihood~~ of having co-existence compared to
364 children whose mothers had ~~ve~~ no formal education, whose mother's participation in decision-making
365 in the family was ~~3.437%~~ (aOR: 0.6663; 95%CI: 0.474 to 0.944) lower ~~oddslikelihood~~ of having co-
366 existence after adjusting for confounding variables. The child whose mother was ~~classified as thin~~
367 ~~underweight~~ (BMI<18 kg/m²) was 1.890 times (95%CI: 1.204 to 2.790) more ~~likely-likely~~ and mother
368 ~~with anemia were 2.43 times more likely to~~ have co-existence compared to a child whose mother had
369 normal nutrition status.

370 Table 7: Factors associated with co-occurrence of undernutrition and anemia among children aged
371 6-59 months

Exposure variables	Anemia only vs Normal			Undernutrition only vs Normal			aOR*
	aOR*	95% CI	p-value	aOR*	95% CI	p-value	
Age of child (ref: 6-11 months chi)							
1-3 years	0.37	0.25 – 0.54	<0.001	3.02	1.32 – 6.91	0.009	0.68
4-5 years	0.16	0.11 – 0.24	<0.001	3.16	1.39 – 7.20	0.006	0.21
Wealth Quintile (ref: Poorest)							
Poorer	0.940.97	0.63 – 1.400.64 to 1.46	0.770.882	0.980.98	0.65 – 1.480.64 to 1.51	0.9280.929	0.40.79
Middle	0.980.99	0.64 – 1.490.64 to 1.53	0.9170.959	0.830.91	0.53 – 1.300.57 to 1.46	0.4120.704	0.610.64
Richer	1.040.92	0.67 – 1.610.59 to 1.45	0.8690.723	0.480.5	0.28 – 0.810.29 to 0.85	0.0060.011	0.580.59
Richest	0.490.45	0.30 – 0.790.27 to 0.75	0.0040.002	0.380.42	0.21 – 0.670.23 to 0.76	0.0010.005	0.410.49
Mother's education (ref: no education)							
Basic	0.870.83	0.60 – 1.270.56 to 1.24	0.4730.368	1.000.94	0.67 – 1.490.62 to 1.44	0.9960.78	0.70.76
Secondary and higher	1.020.97	0.67 – 1.560.62 to 1.52	0.9230.904	0.870.8	0.54 – 1.380.49 to 1.30	0.5430.366	0.510.55
Participation in HH decision (ref: nonparticipation)							
participation	0.690.69	0.51 – 0.930.50 to 0.94	0.0160.018	0.730.75	0.52 – 1.020.53 to 1.07	0.0640.111	0.610.63
Exposure to health programs or TV/radioHealth program exposure (ref: No)							
Exposure	1.041.08	0.77 – 1.420.78 to 1.50	0.7830.635	1.001.08	0.70 – 1.420.75 to 1.56	0.9960.688	0.80.99
Mother's BMI							
Maternal nutrition status (ref: average weight)							
Overweight and obese	1.301.25	0.97 – 1.740.93 to 1.68	0.0760.144	0.890.89	0.63 – 1.280.62 to 1.29	0.5390.54	0.80.86
ThinUnderweight	1.171.16	0.80 – 1.710.78 to 1.74	0.4260.455	1.441.54	0.96 – 2.161.01 to 2.35	0.0750.044	1.80.59

Anemia in mother (ref: No)	1.551.59	1.19 – 2.021.21 to 2.08	0.0010.001	1.411.46	1.04 – 1.891.07 to 1.99	0.0250.017	2.432.41
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Normal: participants having neither undernutrition nor anemia

ref: reference group; *aOLR*: adjusted odds-like ratio; *CI*: confidence interval

Bold represents significance at 95% confidence level

^a adjusted for age, sex of child, father's education, parity, place of residence, ecological belt, mother's age at childbirth, and adjusted for each other

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Discussion:

Our secondary analysis of a national survey assessed to determine revealed significant issues in the prevalence and determinants of factors associated with undernutrition, anemia and their co-existence among Nepalese children in 2022. Overall, the prevalence of undernutrition and anemia was 33.5% and 43.4% respectively. The prevalence of coexistence between undernutrition and anemia was 16.0%, and for the co-existence of stunting, wasting, and underweight with anemia were 12.5%, 4.0% and 9.7% respectively. Undernutrition demonstrated a significant association with household wealth quintile, maternal education, parity, and mother's nutritional status BMI. Anemia was observed found to be associated with both the wealth quintile and the mother's anemia status. The simultaneous occurrence co-existence of undernutrition and anemia exhibited significant associations with wealth quintile, maternal education status, maternal involvement in household decision-making, parity, maternal mother's nutritional status BMI, and the child's anemia status. However, undernutrition, anemia, and their co-occurrence were not found to be associated with maternal exposure to health programs via television and radio.

The prevalence of undernutrition defined as anthropometric failure among 6-59 months children in Nepal was 33.5%, which aligns with the rates observed in other LMICs. For instance, a study in Ethiopia reported a higher prevalence of undernutrition among preschool children (50.8%) and children aged 6-59 months (57.3%) [32] reported higher prevalence of undernutrition compared to the findings from our study where we found prevalence of undernutrition among 6-59 months children in Nepal to be 33.5%. Our findings indicate that Nepal's burden, though lower, remains substantial. These differences may reflect variations in economic development, food security, and maternal and child health interventions across countries [33].

The prevalence of anemia among 6-59 months children from our study (43.5%) is higher than the global prevalence (39.8%) but lower than the prevalence in African children (60.2%) [12]. Our study also demonstrates a 9.1 percentage-point reduction in anemia prevalence between 2016 (52.6%) [34] and 2022, indicating progress in addressing this issue. Comparable rates have been observed in neighboring countries. A meta-analysis conducted by Kundu et al in Bangladesh showed a prevalence of 46.8% [35] which was similar to our finding. The latest national family health survey (2019-21) in India reported 67.1% prevalence of anemia [36] which is more than what we found in Nepal. The decline in anemia prevalence in Nepal [23] is encouraging but underscores the need for sustained efforts to address this condition, particularly among vulnerable populations.

A systematic review reported the pooled prevalence of wasting-anemia and stunting-anemia in least-developed countries to be 5.4% and 19.5%, respectively [37]. These rates are higher than our findings from Nepal, a low middle income country, which were 4.0% and 12.5%, respectively.

The high ~~er~~ prevalence of co-existence of undernutrition and anemia in our study (16%) suggests that affected children are trapped in a vicious cycle of micronutrients. Undernutrition impairs iron absorption and utilization, while anemia exacerbates undernutrition by reducing the body's ability to effectively utilize nutrients.

~~Although undernutrition and anemia share several common risk factors, our analysis aligns with prior studies suggesting that their coexistence is more a function of their respective prevalence rather than shared determinants[35]. Similar to a study done by Gosdin we find no association between anemia and undernutrition. Our study suggests that interventions that address the determinants of undernutrition should not be assumed to address undernutrition.~~

The children from the richest wealth quintile had ~~ve~~ lower odds of having co-existence of anemia and undernutrition compared to the poorest wealth quintile. This wealth-related equity gap in undernutrition among children aged 6-59 is attributed to the economic status of households which affects the capability for food insecurity and health care utilization whenever the child is ill [38–40].

The children from a family in which mothers participate in decision-making and mothers who had ~~ve~~ secondary or higher education were less likely to have co-existence of anemia and undernutrition and anemia alone compared to ~~normal~~ children ~~without anemia and undernutrition~~. The health and nutrition of the child highly ~~depends~~ depend on the mother, being the primary caretaker of the child with awareness of the child's health. The analysis of DHS data from Nepal and India suggests that women's decision-making authority (maternal autonomy in household decision-making) improves the nutritional status of children and ~~reduces~~ reduces child mortality, even after controlling for education and wealth[41].

Strengths and Limitations

This study has several strengths. First, it utilized complex survey analysis to address complex survey design and non-response rates. Second, the findings from this study are generalizable to children aged 6-59 months in Nepal and to children of the same age group in demographically and economically similar countries, as it used a nationally representative dataset.

This study is not free from limitations. One of the limitations is due to the missing data we could not utilize some of the important dietary factors like minimum dietary diversity score and birth weight of child. Second, due to the cross-sectional nature of the study, the directionality of association is not possible to determine. ~~Lastly~~ Third, the results from this study may not be generalizable to populations outside Nepal. Lastly the wide confidence intervals observed in the associations between children's

447 nutritional status and anemia suggest limited statistical power, likely due to small sample sizes within
448 some subgroups.

450 Conclusion

451 The prevalence of undernutrition, anemia, and their co-existence remains high among children aged 6-
452 59 months in Nepal. ~~Anemia and undernutrition were not found to be associated with each other.~~
453 ~~However,~~ The co-existence of anemia and undernutrition was less likely among children from ~~richest~~
454 ~~highest~~ wealth quintile, those whose mothers ~~who~~ attained at least secondary level education, and ~~those~~
455 ~~whose mothers~~ participated in household decision-making. Conversely, co-existence was more
456 prevalent among children whose mothers had poor maternal nutritional status. It is essential to
457 implement holistic programs and targeted interventions to empower poor families and women from
458 high-risk districts ~~thereby to~~ improving the ~~nutritional status of~~ children's health.

460 Data availability

461 We used publicly available data from Nepal Demographic and health survey database upon request
462 from the official website of "The DHS program" [23]. The link to database is:
463 https://dhsprogram.com/data/dataset/Nepal_Standard-DHS_2022.cfm?flag=1

465 Conflict of interest

466 The authors declare no conflicts of interest.

468 Funding

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607 Tables

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608 Table 1: List of exposure variables with their definition

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609 Table 2: Characteristics of 6-59 months children

610 Table 3: Co-existence of anemia and undernutrition conditions (n=2,335)

611 Table 4: Distribution of undernutrition, anemia and their coexistence across exposure variables

612 Table 5: Factors associated with undernutrition among children aged 6-59 months

613 Table 6: Factors associated with anemia among children aged 6-59 months

614 Table 7: Factors associated with co-occurrence of undernutrition and anemia among children
615 aged 6-59 months

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617 Figures

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618 Figure 1: Prevalence of coexistence of different forms of undernutrition among A) children aged
619 6-12 months, B) children aged 1-3 years, C) children aged 4-5 years and D) all children aged 6
620 months to 5 years.

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621 Figure 2: Prevalence of anemia among A) children aged 6-11 months, B) children aged 1-3 years,
622 C) children aged 4-5 years and D) all children aged 6 months to 5 years.

623 Figure 3: Prevalence of co-existence of anemia with stunting, wasting, underweight, and overall
624 undernutrition

625 Figure 4: Distribution of anemia, overall undernutrition and their co-existence of anemia across
626 exposure variables

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628 *Figure 1: Prevalence of stunting, wasting, underweight, undernutrition and anemia among 6-59 months children of*
629 *Nepal*

630 *Figure 2: Prevalence of co-existence of different combination of stunting, wasting, underweight, or undernutrition with*
631 *anemia among 6-59 months children of Nepal*

632 *Figure 3: Distribution of anemia, undernutrition and co-existence by wealth quintile, mother's health program exposure,*
633 *Mother's education and mother's participation in household decision-making*

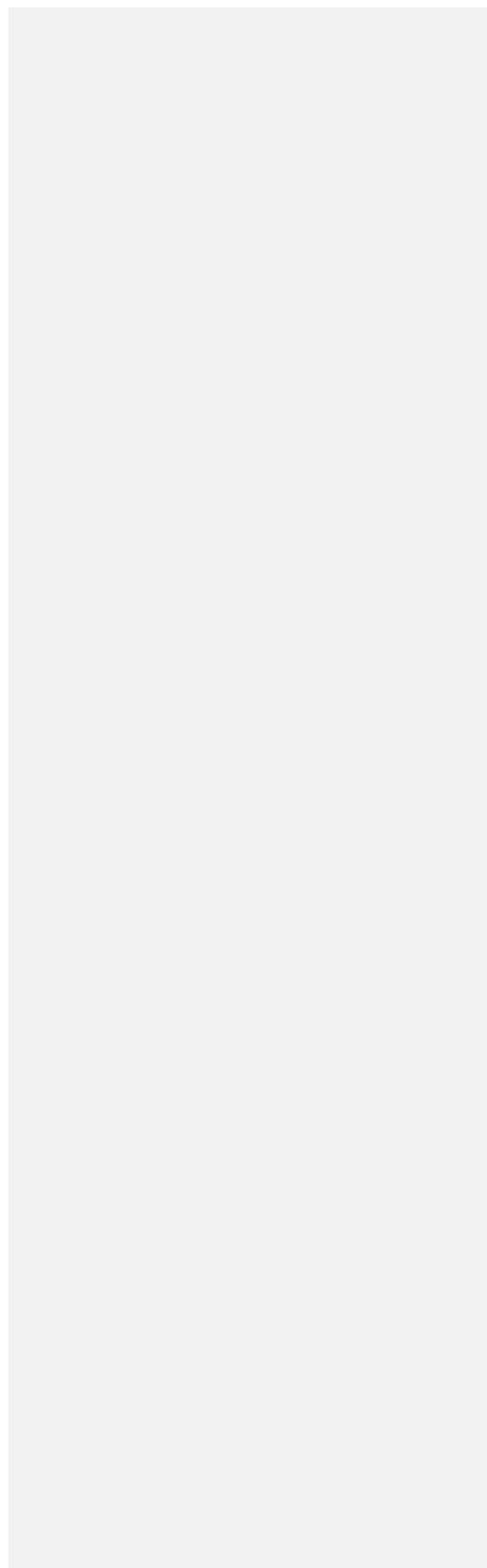
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Dear reviewers and editor,

Thank you for your time and effort to review and provide insightful and constructive comments on our research paper. We have addressed the comments point-by-point as follows:

Reviewer #1:

1. In the methods, authors state "we will analyze the data of 2,395 children from the NDHS 2022 dataset". However, the Abstract and the Results show 2,335. This discrepancy must be clarified. The authors should state the initial sample size and provide a clear explanation for any exclusions.

Response: Both numbers are corrected, 2335 is weighted frequency and 2395 is unweighted frequency. We have corrected and specified in the current version of the manuscript. In line 36, line 139, and table 2 (2nd and 3rd row) are now clarified the weighted and unweighted frequencies.

2. In Table 4, in the factors associated with anemia, the authors include "Nutritional status of child" as an independent variable. Given the cross-sectional nature of the data and the study's aim to assess coexistence, this approach may be problematic as it implies a causal direction that cannot be established and introduces potential endogeneity... The association between undernutrition and anemia is more appropriately explored through the multinomial logistic regression model (Table 5). In my opinion, this variable should be removed from the binary logistic regression model for anemia to avoid confusion and circular reasoning.

Response: Thank you for feedback, I agree with your comment, and we have removed under-nutrition and anemia as independent variables and treated them as dependent variables (Table 4, 5, and 6).

3. The manuscript uses the term "adjusted odds-like ratio (aOLR)". To the best of my knowledge, the output of a multinomial logistic regression model is typically referred to as a relative risk ratio (RRR) or, if presented as odds, as an odds ratio (OR). Please consider using a standard term and define it clearly. For example, "adjusted odds ratios (aOR) from the multinomial logistic regression model, with 'Normal' as the reference outcome category."

Response: The previous term was taken from an old textbook. Now we use an adjusted odds ratio (aOR) term for the multinomial logistic regression model to be consistent with the current terminology. The aOLR was now replaced with aOR throughout the paper.

4. The abstract states that children with malnourished mothers had a "90% (95% CI:1.24 to 2.90) higher likelihood" of coexistence. The CI corresponds to an OR of 1.90 (as shown in Table 5, which represents a 90% increase in odds, not likelihood. This distinction is important, and the language should be precise throughout the manuscript. The abstract should be revised to state: "...children with malnourished mothers had 90% higher odds (aOR: 1.90; 95% CI: 1.24 to 2.90) of..."

Response: Thank you for the feedback. We have corrected this throughout the manuscript. In line 41, Line 290-301, and other lines, we corrected sentences as per your comment.

5. Please correct table numbers, as they should be sequential (there are two tables numbered 1); perhaps using a section numbering?

Response: Thank you for this feedback. We have corrected table numbers throughout the paper.

Minor Comments

5. In the abstract, I suggest breaking down the results section into two or three separate sentences, for improved readability.

Response: The sentence (line 36-42) is split into 2 sentences.

6. In the introduction, the rationale for the study is clear. However, the statement that "there are limited studies exploring factors associated with the co-existence of undernutrition and anemia among U-5 children in Nepal" could be strengthened by briefly mentioning what the few existing studies (if any) have found or what specific gaps remain.

Response: Few existing studies and their findings have been mentioned in the introduction along with the specific gap remaining.

7. In the methods, the definition of "undernutrition" is based on the Composite Index of Anthropometric Failure (CIAF) concept. It would be beneficial to explicitly name this index, as it is a well-established term and is supported by the cited reference.

Response: Thanks. We have clarified this term.

8. The authors state they removed the 'province' variable from regression models due to a variance inflation factor (VIF) > 2. A threshold of 2 is quite conservative; VIF thresholds of 5 or 10 are more commonly used. Please provide justification for this choice.

Response: The VIF for the province variable was 13.31, so it was removed from the model based on either criterion. Note that the information of provinces is captured by the region and place of residence variable, i.e., these two variables were highly collinear.

9. The manuscript should explicitly state how missing data was handled in the statistical analysis. Several variables in Table 1 (e.g., father's education, mother's age, parity) have a substantial number of missing values. Were these cases excluded via listwise deletion for each model? This should be clarified in Statistical Analysis subsection.

Response: They were excluded during analysis. Now it is mentioned in the statistical analysis subsection

10. The manuscript would benefit from a thorough proofread to correct minor grammatical errors and typos. For example: "presence anemia in children" (pg. 18, at the bottom) should be "presence of anemia"; "...those whose mothers who attained at least secondary level education, and those whose mothers participated..." is repetitive.

Response: Thank you for your feedback. We wrote this manuscript without relying on any AI tools for grammatic check. We have thoroughly proofread the manuscript manually and corrected all the grammatical errors seen in the paper.

Reviewer #2: The topic presented in the manuscript is highly relevant for public health in developing countries.

Please consider making the title more specific to the scope and methods of your research.

The definition of undernutrition is vague: "Stunting, wasting, underweight or any combination" — does this mean a child is considered undernourished if they have any one of the three? Or only if they have multiple?

Response: The child was considered to have undernutrition if they had any one of the three conditions. This statement is clarified in line 163, page 6.

Odds-like ratio" (aOLR) is uncommon and may confuse readers. Standardize with “adjusted odds ratio (aOR)” unless you are referring to a specific statistical nuance that requires clarification.

Response: Thanks. This has been corrected throughout.

Review grammatical issues and ambiguity of same phrases. Some sentences are wordy or unclear.

Response: we have proofread the manuscript again and corrected all recognized grammatic errors.

Contradictory finding needs explanation: You report that anemia and undernutrition were not associated with each other, yet some children had both. This warrants further discussion. If they co-exist at 16%, why is the statistical association not significant?

Response: Thank you for your comment. We hypothesized the existence of an association between undernutrition and anemia after multivariate adjustment. However, our data did not support this hypothesis, thus the multivariable analyses (Table 5 and 6) were conducted with undernutrition and anemia separately. The crude association between undernutrition and anemia may be due to confounding factors such as socioeconomic status and regions included in the models. We also add this point in the discussion section.

In conclusions “Targeted interventions” is vague. What kind of interventions? Nutrition-specific? Maternal education? Social protection? Consider suggesting specific policies or areas for programmatic focus.

Response: Thank you for this comment. We have added specific interventions suggesting specific policies and areas of programmatic focus. That is, policies will be tailored to the specific needs in regions and based on the socioeconomic status of the people.

Reviewer #3: This is an interesting and well-written paper on a relevant topic. Please find below some issues which should be addressed to further improve the manuscript.

General remarks:

- The authors might re-check their article according to the STROBE checklist. For example, the study's design should be indicated with a commonly used term in the title or the abstract.

Response: We have clarified in the abstract that this is a secondary data analysis of a cross-sectional survey.

- In the spirit of Open and Reproducible Science, the analysis code should be made available in an online repository together with a data dictionary, and the respective URL should be mentioned in the Methods section.

Response: The data and analysis code are accessible in the GitHub
<https://github.com/Adrisyaone/Undernutritiona-and-anemia>

Detailed points:

- The title says this was a secondary analysis of the NDHS data. However, this term is not mentioned in the main text. It would therefore be helpful to give some context to the NDHS study, e.g. what its original aim was, how many participants were included etc.

Response: We have added original aim and number of participants for the original NDHS survey (line 127, page 4).

- Abstract: "were ... lower likelihood" is grammatically not correct. Please revise.

Response: Thank you for your feedback. It is now corrected as "lower odds".

- Is "NDHS" the correct abbreviation for "Nepal Health Demographic Survey"? "NHDS" would seem more appropriate and is also mentioned like this once in the Methods section.

Response: NDHS is Nepal Demographic and Health Survey, so NDHS is the common abbreviation.

- The description of Nepal as a country might fit better into the Introduction. It would be helpful to the Reader to put the HDI values into context, e.g. compare them with those of the highest and least developed countries.

Response: Thanks. We have clarified this point in the method section regarding Nepal's HDI and Global Hunger Index.

- "In this study, we will analyze the data of 2,395 children from the NDHS 2022 dataset." It should be mentioned for clarity that all these children were U-5. Were these children all U-5 children included in the DHS 2022 data? If not, by which criteria were the children selected for analysis?

Response: In this study all Under-5 children aged 6 months and above were taken for analysis. Now, we have mentioned which subsets of under 5 children are included for analysis.

- Please add a paragraph explaining how the data were collected, e.g. how height, weight and blood levels were in children and mothers measured, by whom (e.g. trained physicians) and in which setting (at school, in the children's homes...)? How were the remaining variables from table 1 and the potential confounders measured?

Response: we added brief information on these measures (Table 1) and referred them to the official documents of NDHS.

- I think "mother's nutritional status" is not a suitable word for describing their BMI category.

Response: it is changed to "Mother's BMI" throughout the manuscript

- Which population was used as reference to calculate z-scores of heights and weight?

Response: They were internally calculated (based on the overall population) in NDHS 2022.

- "province variable" is not a common statistical term. Please revise.

Response: Thank you. It is revised.

- "10.5% were 6-12 months old": Table 1 says 10.4%.

Response: Thank you. It is corrected

- Please note that there are two tables labelled as table 1.

Response: all tables are re-labeled and correctly numbered.

- The VENN diagram in Figure 1 is quite nice, but I think the information in Figure 2 might better be presented in a table, together with exact numbers (not only %). In Figure 2, it is not clear what the percentages in the upper right of each of the four panels indicate.

Response: The figure 1 is splitted into multiple Venn diagram based on the age group of children.

- Also in table 2, exact numbers should be mentioned in addition to %.

Response: Currently, the overall weighted N and unweighted N is kept in the table. This may make the table easy to read and the exact numbers could be computed in ad hoc.

- Figure 3: There should be no interpretation in the figure-legend, so the last sentence in the legend should be omitted. Instead, the meaning of the vertical bars (95% CIs?) should be explained.

Response: Thanks. Fixed. We explained the figure in the notes and text instead of legend.

- Were the analyses presented in tables 3 and 4 mutually adjusted for each other variable shown in the table? If not, why not?

Response: No, they were not mutually adjusted for each other. Both undernutrition and anemia were treated as dependent variables. The tables were revised as Table 5 and 6.

- "The child whose mother was thin" is awkward English wording and should be revised.

Response: The statement was rephrased as “The child whose mother was classified as thin (BMI<18 kg/m²)....”.

- The second p-value column is incorrectly assigned to "Coexistence vs Normal".

Response: This issue with the table is corrected.

- In general, I would suggest removing all p-values from the tables, as the 95% CIs already contain information on whether an association was statistically significant or not.

Response: We agree with your statement, but some readers prefer p-values, thus we kept them in the table.

- Discussion, first sentence: The prevalence of undernutrition should also be explicitly mentioned in the results section. The anemia prevalence of 43.5% is reported as 43.4% on other places in the manuscript.

Response: it was a typo which is now corrected

- "...which aligns with the rates observed in other LMICs. For instance, a study in Ethiopia reported a higher prevalence..." These two sentences do not fit together in this way.

Response: Thank you for the feedback, we have changed the sentences and write in this way:

A study in Ethiopia reported a higher prevalence of undernutrition among preschool children (50.8%) and children aged 6-59 months (57.3%) [32] reported higher prevalence of undernutrition compared to the findings from our study where we found prevalence of undernutrition among 6-59 months children in Nepal to be 33.5%.

- "The decline in anemia prevalence in Nepal...": Does this mean this prevalence was higher previously? If so, a reference should be given which supports this statement.

Response: Thank you, I've added reference to the statement.

- "...our findings from Nepal, a low-middle-income country," This does not need to be repeated again.

Response: Corrected

- "The higher prevalence of co-existence of undernutrition and anemia in our study...": Higher compared to what?

Response: The term higher was changed to high and this sentence wants to say 16% is high.

- "Our study suggests that interventions that address the determinants of undernutrition should not be assumed to address undernutrition." I do not understand this sentence. Should the second "undernutrition" rather say "anemia"?

Response: Thank you for the feedback. The last word should be anemia but because of typo, this sentence is not clear I've corrected this sentence.

- "The children from the richest wealth quintile have lower odds..." Throughout the manuscript, when relating to results from this study, past tense should be used.

Response: Corrected

- How were non-response rates addressed, and how high were these rates (overall and for U-5 children)?

Response: This is the complex survey analysis, which addresses the non-response rates and allows for generalization throughout the population. The response rates were higher which was >95%.

- As a further limitation, the authors should discuss that the CIs between children's nutritional status and anemia were quite large, indicating that the statistical power / sample size was limited with respect to these associations.

Response: Thank you for the constructive feedback, we have added this to the limitation section.